When Selling Spreads: Local Decisions, Limited Resources, and Market Crashes in a Simulated Market*

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Abstract

Changes in investor decisions can spread through a market, but the resulting price decline depends on how trades execute. We developed a market model in which traders inside a grid decided whether to buy, hold, or sell from their neighbors' decisions, while technical traders at selected positions along the edge of the grid responded to price. Paired markets shared the same history until selected traders in one market received a lasting change in their local decision rule, while the other market, a no-switch control, kept the original rule. A Sell majority formed once enough traders received the new response, even though every other trader kept its original rule. When the technical traders were held so that no decision responded to the price, majorities still formed, so responses to neighbors alone could produce widespread selling. Comparisons across market sizes and observation periods located a crossover whose position continued to move with the time observed. To connect decisions to prices, we cleared orders against a dealer whose finite cash and share reserves kept a constant product. Those reserves set both the execution price and the next quoted price, so a round trip that returned the dealer's shares could not extract its cash. Shallow positive reserves accompanied severe declines while rejecting few orders that traders had the resources to complete, whereas zero dealer depth allowed only trades between traders at a fixed price. Markets with shallow reserves also had large drawdowns without the rule switch, so the effect of the switch could be measured only against the paired controls. Changing trader cash and shares altered later decisions only when technical traders could respond to the resulting prices. Transaction fees did not reduce price declines in every tested condition. The model thus separated the spread of selling decisions from the severity of a simulated crash. Limited trading resources did not generally protect against severe declines, because their financial consequences depended on how execution set the price.

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

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1 Introduction

Financial markets are the collective outcome of human interactions. They reflect both our ingenuity and our failings. In its 2011 report on the 2008 financial crisis, the Financial Crisis Inquiry Commission paraphrased Shakespeare: "the fault lies not in the stars, but in us" [1, p. xvii]. In markets for traded financial assets such as stocks, investors make decisions with different information, expectations, and financial resources, and they respond to prices and to the actions of others when deciding whether to buy, hold, or sell. An investor's decision to sell can prompt others to do the same. Sellers need buyers who are willing and able to purchase their assets, and the prices buyers will pay reflect their views of the assets' value and the funds they can commit to the purchase. If sellers cannot find buyers at the current price, they may accept lower prices to complete their sales, and the resulting price changes can then influence investors' later decisions. Under what conditions can a change in the behavior of some investors lead to widespread selling, and when does that selling produce a market crash?

Agent-based models study how individual trading decisions and interactions among investors produce market behavior. This literature includes many approaches to learning, imitation, and price formation [2, 3], and its models differ in the information available to traders and in how their trades affect prices. Farmer and Joshi studied traders who followed price trends or compared prices with their estimates of asset value. A market maker absorbed their net orders and adjusted the price, and the resulting price changes then influenced later trading decisions [4]. In the Santa Fe artificial stock market, traders learned forecasting rules from market prices and dividends and used those forecasts to guide their trades [5]. In previous work, we developed an agent-based market for trading multiple assets, in which agents submitted orders to a continuous double auction that matched compatible buy and sell orders as they arrived. Traders with different decision rules therefore interacted through a common trading mechanism [6]. Alongside interactions through prices and trades, herding models describe how traders respond to the behavior of others [7–9]. Arranging traders in rows and columns provides a simple representation of the connections through which influence operates: each position represents a trader, and neighboring positions identify the traders whose decisions it can observe. The information passed through these connections can be as simple as a trader's current decision to buy or sell [10], and a change in one trader's decision can affect its neighbors, whose responses can then influence other traders. Models inspired by the Ising model use these local interactions to represent imitation. In Bornholdt's simplified model, traders tended to follow their neighbors, but a growing imbalance between buyers and sellers across the market increased their tendency to reverse their current decisions [11]. Iori used a related model in which traders could buy, sell, or remain inactive, repeatedly exchanged their current decisions with their neighbors, and used the resulting signals to revise their own choices [10]. These models used similar arrangements of traders but different rules for responding to neighboring decisions. The arrangement specifies who can influence whom, while the decision rule specifies how each trader responds.

Cellular automata use local rules to update traders' decisions in discrete steps [12, 13]. Wolfram arranged traders on a line, where each trader responded to its two neighbors and the cumulative difference between buying and selling served as a price analogue [14]. Later models combined cellular automata with trader balance sheets [15] or used stochastic rules for interactions between neighboring traders [16–20]. Our model represents a synthetic market with traders arranged in rows and columns. Two types of traders occupy the market interior, and both use their four neighbors' intended actions to decide whether they want to buy, hold, or sell. Their rules differ in a single response to their neighbors' intended actions, and we vary the relative numbers of the two types to study how the mixture affects the spread of selling intentions. Technical traders along the edge of the synthetic market base their decisions on prices, and the interior traders next to them respond to those decisions. The financial consequences of intended trades depend on the resources available for exchanging cash and shares, and earlier agent-based models have examined limits on traders' cash and asset holdings and on their ability to borrow [21, 22]. In our model, Buy and Sell indicate intended actions, whereas a completed trade transfers shares from a seller to a buyer and cash from the buyer to the seller. Traders cannot borrow cash or sell shares they do not own. A trader without shares can still indicate a desire to sell but cannot carry out the sale, and a trader without enough cash can indicate a desire to buy but cannot carry out the purchase. Because neighboring traders respond to intended actions even when no trade occurs, selling intentions can spread without a corresponding increase in completed sales. A dealer acts as the market maker, using its finite cash and share reserves to complete trades that cannot be matched between traders. Completed trades preserve the product

of the dealer's cash and share reserves, following a constant-product market-making approach [23], and after each round of trading the dealer's quoted price per share equals its remaining trading cash divided by the number of shares it holds. When the dealer buys shares from traders, it has less cash and more shares, so dividing the smaller cash balance by the larger number of shares gives a lower quoted price. The size of the dealer's reserves therefore determines both which trades can be completed and how much the quoted price changes when the dealer buys or sells shares. Technical traders use the dealer's updated quoted price to make their next decisions, which then influence neighboring interior traders.

In this study, we constructed a synthetic market based on a three-state cellular automaton to examine whether mixing two trader types could lead to widespread selling intentions, and how trader holdings and dealer reserves affected the accompanying price decline. In each paired rule-switch comparison, both markets had the same number of traders and shared a period of trading with only the original trader type in the interior. We then assigned the second decision rule to a chosen fraction of interior traders in one market, while the other market kept the original rule. Trading intentions, trader holdings, dealer reserves, and prices matched at the rule switch. The traders could then make different decisions and complete different trades. We varied the fraction of interior traders using the second rule to determine when at least half of the interior traders wanted to sell, which we called a Sell majority. In separate experiments, we varied the total number of traders and the simulation duration to examine how market size and time affected the spread of selling. We also ran markets with all technical traders held in Hold to test whether a Sell majority could develop when no decision responded to the price. Interior traders continued to respond to neighboring intentions and could complete trades with one another or the dealer, so trades with the dealer could still change its reserves and quoted price. In the rule-switch experiments, keeping technical traders in Hold prevented price changes from influencing interior traders' intentions, but prices still affected which trades could be completed. Buyers needed enough cash to purchase shares, sellers needed shares to sell, and each sale required a buyer or dealer able to pay the execution price. We varied dealer reserves and traders' cash and shares separately to examine how those resources affected completed trades and price declines. We also examined crowding by having groups of traders follow shared advice or copy another trader's completed trades. We varied the fraction of followers and the number of sources to examine effects on other traders' decisions, prices, and the distribution of gains and losses. For shared advice, we tested whether followers' influence on neighboring traders contributed to their financial gains. Together, the rule-switch and following experiments showed how individual trading decisions could spread through a market, affect prices, and redistribute cash among traders.

2 Methods

2.1 Three-state cellular automaton

We represented the synthetic market as L rows and L columns, with $(L-2)^2$ traders in the interior, and recorded each trader's intended action as Buy ($s=0$), Hold ($s=1$), or Sell ($s=2$). Technical traders occupied selected positions along the edge and formed their intentions from the dealer's quoted price and recent price history, while the remaining edge positions stayed in Hold. Each interior trader formed its local decision from the intentions of its four neighbors above, below, to the left, and to the right. The decision rules were totalistic: each trader's next intention depended only on the sum of these four neighboring states [13], so for interior traders next to the edge this sum could include the intentions of technical traders. After trades settled at each step t, the technical traders updated their intentions using the new quoted price and price history. All interior traders then updated together using the previous interior states and the newly updated edge states, producing intentions for the next step's orders. We denoted the four neighbors of trader i by ∂i and the state supplied by neighbor j at step t by $s_j^{\text{in}}(t)$, and the totalistic rule used their sum $c_i(t)$, defined by:

$$c_i(t) = \sum_{j \in \partial i} s_j^{\text{in}}(t) \in \{0, 1, \ldots, 8\}. \tag{1}$$

Each totalistic decision rule assigned an intended action to each of the nine possible neighbor sums. For a sum $k \in \{0, 1, \ldots, 8\}$, we denoted the trader's next intended action by $a_k \in \{0, 1, 2\}$, using the codes for Buy, Hold, and Sell. Specifying these nine outputs defined the complete rule, giving $3^9 = 19{,}683$ possible rules for the four-neighbor model. We identified each rule by its Wolfram rule

number R, with the outputs written as base-three digits from a_8 on the left to a_0 on the right [12, 13], so the rule number was defined by:

$$R = \sum_{k=0}^{8} a_k 3^k. \tag{2}$$

We used Rule 19280, $(222110002)_3$, as the baseline and Rule 19286, $(222110022)_3$, for the selected traders after the rule switch. The two rules differed only in their response to neighbor sum $k = 1$, corresponding to three neighbors intending to Buy and one intending to Hold. For this configuration Rule 19280 assigned Buy and Rule 19286 assigned Sell as the trader's next intended action, whereas all other neighbor sums produced the same intended action under both rules. Before each run, we specified an approximate fraction f of interior traders to receive Rule 19286 later in the simulation and selected $\mathrm{round}(f(L-2)^2)$ traders uniformly without replacement. We initialized the interior with $\mathrm{round}(0.05(L-2)^2)$ traders intending to Buy and the same number intending to Sell, at positions chosen uniformly without replacement, and all remaining interior traders began in Hold. All interior traders initially used Rule 19280, allowing the market to develop before the scheduled rule switch. At the switch, the selected traders received Rule 19286 and first used it to determine their next intended action, while all other interior traders continued using Rule 19280. Assigning the new rule did not execute a trade or reset any trader's cash or shares.

2.2 Orders, trader resources, and completed trades

A trader's intended action generated a desired order, but completing the order required cash or shares and a counterparty. Each interior trader and technical trader held cash w_i and an integer number of shares q_i. A Buy action requested one share, a Sell action offered one share, and a Hold action generated no order. All completed trades within a step used the same base execution price, and a buy required enough cash to pay the execution price and any transaction fee, whereas a sale required at least one share. We allowed sales below a trader's average purchase price $\bar{c}_i$, so the model did not impose the reluctance to realize losses described in behavioral studies [24, 25]. We classified desired orders that could not be completed because the trader lacked cash or shares as *trader suppression*. We allowed traders' Buy and Sell intentions to influence their neighbors even when the traders lacked the resources to carry them out, so cash and share holdings limited completed trades but did not determine which intentions entered neighboring decisions.

The dealer acted as market maker for orders unmatched between traders. Clearing fixed the net trade direction, selected its largest feasible size and price, then allocated orders at that price. For N_{tr} interior and technical traders, dealer depth h, the dealer's initial shares per trader, set initial shares $H = \mathrm{round}(hN_{\mathrm{tr}})$ and cash $P_0 H$, with initial price $P_0 = 100$. The dealer could not borrow and, for positive reserves, retained at least one share while preserving the product K of trading cash and shares [23]. At step t, the starting reserves were cash W_{t-1} and shares Q_{t-1}. A proposed net purchase x from the dealer was positive for trader buys and negative for trader sales. Its base execution price $F_t(x)$ had to preserve the reserve product after the dealer exchanged x shares for cash $F_t(x)x$, according to:

$$K = W_{t-1}Q_{t-1},$$
$$[W_{t-1} + F_t(x)x][Q_{t-1} - x] = K.$$

For a nonzero proposed net trade, the corresponding price was:

$$F_t(x) = \frac{W_{t-1}}{Q_{t-1} - x}, \qquad Q_{t-1} - x \geq 1. \tag{3}$$

For no net trade, we used $F_t(0) = W_{t-1}/Q_{t-1}$ to match opposing orders without changing dealer reserves. The fee fraction of each completed order's base price was τ, with $\tau = 0$ except in fee comparisons. At each proposed price $F_t(x)$, the eligible counts were $B_t(x)$ Buy orders backed by cash $(1+\tau)F_t(x)$ and $S_t(x)$ Sell orders backed by at least one share. First, we fixed the trade direction using eligible orders at $F_t(0)$, the price for no net trade. The direction d_t was positive for net purchases from the dealer, negative for net sales, and zero for no net trade, as given by:

$$d_t = \mathrm{sign}\{B_t(0) - S_t(0)\}.$$

Next, with a nonzero direction fixed, we chose the largest nonnegative integer size k that left the dealer at least one share and was supported by the excess eligible orders at its own proposed price. The selected net purchase x_t was therefore:

$$x_t = d_t \max\left\{k \in \mathbb{Z}_{\geq 0} : Q_{t-1} - d_t k \geq 1, \quad d_t[B_t(d_t k) - S_t(d_t k)] \geq k\right\}. \tag{4}$$

We set $x_t = 0$ when $d_t = 0$. Finally, we allocated orders at the selected price $F_t(x_t)$, fixing the eligible counts at $B_t = B_t(x_t)$ and $S_t = S_t(x_t)$. The completed trader buys n_t^B and sells n_t^S were given by:

$$n_t^B = \min\{B_t, S_t + x_t\}, \qquad n_t^S = n_t^B - x_t. \tag{5}$$

All eligible orders on the smaller side completed. Orders on a rationed side were ranked by a reproducible score using simulation seed, step, trader identity, and side, with identity breaking ties. Eligible orders left unfilled were *dealer rejection*. Rejection could reflect the minimum share reserve or a price at which eligible orders no longer supported a larger net trade, without exhausting dealer cash. The selected net purchase per trader ϕ_t summarized completed orders without separately setting price:

$$\phi_t = \frac{x_t}{N_{\mathrm{tr}}}. \tag{6}$$

All completed orders used the common base execution price $F_t = F_t(x_t)$, so the dealer's remaining shares Q_t and trading cash W_t were:

$$Q_t = Q_{t-1} - x_t, \qquad W_t = W_{t-1} + F_t x_t. \tag{7}$$

After settlement, we set the dealer's quoted price P_t equal to its remaining trading cash divided by its remaining shares, and substituting the base execution price into these balances gave:

$$P_t = \frac{W_t}{Q_t} = P_{t-1}\left(\frac{Q_{t-1}}{Q_{t-1} - x_t}\right)^2. \tag{8}$$

This quoted price supplied the price signal used by technical traders and differed from the base execution price whenever $x_t \neq 0$. For each completed order, a buyer paid $(1+\tau)F_t$ and a seller received $(1-\tau)F_t$, and the dealer collected the fees in a separate cash account G_t, initialized at zero and excluded from the trading reserves used to set prices:

$$G_t = G_{t-1} + \tau F_t(n_t^B + n_t^S), \qquad G_0 = 0. \tag{9}$$

No fee was charged on an unfilled order. Cash was conserved across traders, dealer trading reserves, and the fee account, except at scheduled changes in trader resources, and shares were conserved across traders and the dealer, except at scheduled share interventions. In exact arithmetic, the constant reserve product $W_t Q_t = K$ meant that an isolated sequence of trades returning the dealer to its initial share reserve also restored its initial trading cash. Such a round trip could not extract net cash from the dealer and, at positive τ, paid fees to the dealer, although this result did not rule out gains from trading with other traders. Dealer reserves determined both the funding available for trades and the response of the quoted price to net trading. The comparison with large reserves used dealer depth $h = 100$, whereas at dealer depth $h = 0$ the dealer held no trading cash or shares. Only opposing trader orders backed by sufficient resources could then be matched, the price remained at P_0, and fees collected in this case did not fund dealer trading. This separate case therefore provided a fixed-price control (Algorithm S2).

2.3 Technical traders and price feedback

Technical traders brought changes in the dealer's quoted price into neighboring decisions. We selected one quarter of the edge positions uniformly without replacement for these traders, while the remaining edge positions stayed in Hold. The technical traders followed a contrarian strategy of the Bollinger-band type, in which each technical trader compared the quoted price with a band around a recent price average. A price below the lower limit generated a Buy intention, a price above the upper limit generated a Sell intention, and prices within the band, including its limits, generated Hold. These intentions influenced neighboring interior traders and supplied orders for the next trading step. This contrarian response contrasted with the trend-following moving-average rules often tested in market data [26]. The reference price μ_t was an exponentially weighted moving average of the dealer's quoted price, and the band width used σ_t, an exponentially smoothed absolute departure from this average rather than a standard deviation. Both quantities used the smoothing parameter $\alpha = 1/60$. After settlement, we first updated the average using the new quoted price P_t and then updated the departure:

$$\begin{aligned} \mu_t &= \mu_{t-1} + \alpha(P_t - \mu_{t-1}), \\ \sigma_t &= \sigma_{t-1} + \alpha\big(|P_t - \mu_t| - \sigma_{t-1}\big). \end{aligned} \tag{10}$$

We initialized the average at $\mu_0 = P_0$ and the departure at $\sigma_0 = 0$. Each technical trader received a fixed multiplier $k_j \sim U[0.5, 1.5]$, giving lower and upper limits $\mu_t - k_j\sigma_t$ and $\mu_t + k_j\sigma_t$.

At the start of each step, we applied any scheduled changes to decision rules or trader resources, and traders then submitted orders from their existing intentions. After settlement, technical traders updated their bands and intentions using the dealer's new quoted price, and interior traders then updated simultaneously, using the previous intentions of neighboring interior traders and the new intentions of adjacent technical traders. The resulting intentions supplied orders for the following step (Algorithm S1). To remove price feedback, we kept every technical trader in Hold from initialization, so these traders submitted no orders and communicated Hold to their neighbors. Interior traders could still trade, and cash, shareholdings, and the dealer's quoted price could still change, but those changes could no longer influence interior intentions through the technical traders. We compared this condition, *technical traders held*, with *technical traders active*, in which they followed the band rule. Both conditions began with the same starting market and used the same interior traders selected for the rule switch, so the comparison tested whether a Sell majority could develop from responses to neighbors without feedback from prices. Figure legends abbreviate these conditions as held and active.

2.4 Paired rule-switch experiments

We measured rule-switch effects by pairing switched markets with no-switch controls that retained the original rule. Each pair began from the same starting market and used Rule 19280 for every interior trader for 1,000 steps. At the start of step 1,001, we assigned Rule 19286 to a preselected fraction f of interior traders, who retained their current intentions and financial resources, while the control continued with Rule 19280. The two simulations therefore shared the same history up to the rule switch. Each condition in the four main comparisons used 50 starting markets (Supplementary Table S1). Every trader initially held 1,000 cash units and ten shares, and each simulation continued for 3,000 steps after the baseline. Additional comparisons tested other decision rules and a one-time instruction to sell, and we also kept the technical traders' band widths fixed after the rule switch, tested additional dealer depths, and extended the observation period. The supplementary protocols give the conditions and parameter values for these comparisons (Appendix A.5).

We varied the fraction of traders given the new rule, market size, and financial resources to compare the spread of Sell intentions with changes in completed trades and prices. The rule-switch comparison used a 64×64 grid, active technical traders, and dealer depth $h = 100$ at fractions $f \in \{0.10, 0.15, 0.20, 0.25\}$, whereas the dealer-depth comparison used 11 fractions between 0.10 and 0.25 at eight dealer depths h from 0 to 100, with technical traders active or held. For the size comparison, we used grid sizes $L \in \{32, 48, 64, 96, 128\}$, with technical traders held at dealer depth $h = 100$ or active at $h = 1$ and $h = 100$. At a fixed dealer depth h, initial dealer cash and share reserves increased in proportion to the number of traders. To test how trader resources affected the response to the rule switch, we independently multiplied each trader's current cash and shareholdings by factors $g_w, g_q \in \{0.1, 1, 10\}$ at the switch, which gave nine combinations, including unchanged holdings when both factors were 1. Share counts were rounded to whole numbers, and the intervention did not rescale the dealer's reserves. With other settings fixed, all nine simulations shared the same history before the intervention and assigned the new rule to the same traders. These resource comparisons used switched markets without separate no-switch controls. When technical traders were held, changing resources could alter completed trades and prices but could not alter interior intentions, because neither cash nor shareholdings entered the decision rule. The supplementary material gives the full parameter combinations and verification checks (Appendix A).

We tested fees of 0, 5, 10, and 25 basis points on each completed buy or sell to determine whether trading costs changed the response to the rule switch. One basis point was 0.01% of the base execution price. The rates were chosen for a sensitivity comparison and were not calibrated to a particular market. Fees applied from initialization, so different rates could produce different financial histories before the rule switch. Each condition used 50 starting markets on 64×64 grids, with 1,000 baseline steps and 3,000 subsequent steps. We tested all combinations of fraction $f \in \{0.10, 0.15, 0.175, 0.185, 0.20, 0.25\}$, dealer depth $h \in \{0.1, 1, 100\}$, and technical traders active or held. For each fee rate, dealer depth, and technical-trader condition, one no-switch control per starting market retained the original rule and served all six fractions. Each switched run shared its history with this control up to the switch. This comparison used 8,400 simulations. At each

simulation step, we recorded the interior Sell fraction, the dealer's quoted price, counts of completed orders and buy orders blocked by insufficient cash, total trader cash, and cumulative dealer fee income. We used these histories to identify Sell-majority formation and calculate maximum drawdown after the rule switch. Fee income remained separate from the dealer's trading reserves. We compared each switched run with its no-switch control at the same fee rate. We also compared each positive fee rate with zero fees under the same rule switch. Both comparisons matched simulations by starting market and held other settings fixed. We used 2,000 bootstrap resamples of the 50 matched starting markets to obtain percentile 95% confidence intervals for these differences. Selected comparisons appear in Supplementary Table S12.

2.5 Market outcomes, statistical analysis, and selection of examples

We measured intentions and financial outcomes separately because a Sell majority did not necessarily produce a large price decline. The Sell fraction $m(t)$ was the fraction of interior traders in Sell after the intention update at step t, and a run reached a Sell majority if $m(t) \geq 0.5$ at least once during the observation period after the rule switch. Onset time was the first step after the switch at which this threshold was reached. We estimated majority probability as the fraction of runs reaching the threshold and summarized onset times only for those runs. For dealer depth, market size, and trader-resource comparisons, we also averaged $m(t)$ over the final 500 steps. For the market-size comparison, we denoted this mean by $\bar{m}$ and measured its variation between starting markets using the susceptibility χ_L, the variance of $\bar{m}$ multiplied by the number of interior traders, $(L-2)^2$ [27, 28]. With angle brackets denoting averages over the 50 starting markets at a fixed grid size and parameter combination, we calculated:

$$\chi_L = (L-2)^2 \left[\langle \bar{m}^2 \rangle - \langle \bar{m} \rangle^2\right]. \tag{11}$$

At a fixed grid size, a larger χ_L indicated greater variation in the final mean Sell fraction between starting markets (Eq. 11). For each market size L and condition, we estimated the fractions of traders assigned the new rule at which majority probability reached 25%, 50%, and 75%, denoted by $f_{25}(L)$, $f_{50}(L)$, and $f_{75}(L)$. We used straight lines between probabilities at adjacent tested fractions (linear interpolation), taking the first upward crossing of each probability level and retaining any decreases in the sampled probabilities. The midpoint $f_{50}(L)$ located the response, while a smaller width $f_{75}(L) - f_{25}(L)$ indicated a sharper rise in majority probability. We measured price declines using maximum drawdown, the largest proportional fall from the highest quoted price reached so far within the observation period, including the quoted price immediately before the rule switch. We used price crash to describe a large simulated decline, without a threshold for classifying runs as crashes. We counted orders lacking cash or shares as trader suppression and eligible orders left unfilled as dealer rejection. The reported dealer-rejection fraction was rejected orders divided by eligible orders, using each run's totals after the rule switch, with zero assigned when no orders were eligible. Suppressed and completed order counts covered the same period. We summarized maximum drawdown and dealer-rejection fractions using medians across starting markets under the same conditions. In the fee comparisons, majority effects were differences in probabilities, whereas drawdown effects were medians of the differences within matched pairs. We used 95% Wilson confidence intervals for majority probabilities [29]. Where shown, interquartile bands spanned the 25th to 75th percentiles of outcomes across starting markets under the same conditions.

For the market-size comparison, we added tested fractions to resolve the rise in majority probability, aiming for at least five fractions with estimated majority probabilities strictly between 5% and 95% for each size and condition, allowing up to ten rounds. Each round added the midpoint of every adjacent pair whose lower fraction had a probability below 95% and whose higher fraction had a probability above 5%, using the same 50 starting markets at each added fraction (Supplementary Table S3). For each market size and condition, we drew 2,000 bootstrap samples of 50 starting markets with replacement, keeping the same selection at every tested fraction within a sample. We recalculated f_{25}, f_{50}, f_{75}, the width $f_{75} - f_{25}$, and the maximum susceptibility over tested fractions in each sample, using the 2.5th and 97.5th percentiles for 95% confidence intervals. All requested crossings were bracketed in every bootstrap sample (Supplementary Table S3). To check the sensitivity of these interpolated estimates to the use of straight lines, we fitted a smooth, increasing logistic probability curve and compared its midpoint and width at the same probability levels with the primary estimates. The fitted parameters a_L and b_L controlled the curve's position and steepness, while 0.18 and 0.01 centered and scaled the fraction f for fitting. The fitted Sell-majority probability $p_L(f)$ was given by:

$$p_L(f) = \{1 + \exp\left[-a_L - b_L(f - 0.18)/0.01\right]\}^{-1}, \qquad b_L > 0.$$

We estimated the parameters by maximizing the binomial likelihood of the observed majority counts and refitted the curve in each of the same 2,000 bootstrap samples. Standard errors of its midpoint and width were the standard deviations of their bootstrap estimates (Supplementary Table S10). These intervals and standard errors measured uncertainty from sampling starting markets, conditional on the tested fractions and observation period. They excluded uncertainty from selecting additional fractions and, for interpolated estimates, error from approximating the probability curve by straight lines. A longer observation period would define a different outcome, whose change these uncertainty estimates did not describe.

We selected examples of traders' decisions across the grid from the 3,000-step results, before inspecting images, using proximity to group medians and the lower-numbered starting market to break ties. Selection used 64×64 grids, active technical traders, zero fees, and the baseline resource settings without cash or share interventions. At dealer depth $h = 100$, we selected a run without a majority at fraction $f = 0.15$, both outcomes at $f = 0.175$, and runs with a majority at $f = 0.185$ and 0.20 (Fig. 4). Within each fraction and outcome group, we chose the run nearest the median onset time for runs reaching a majority, or nearest the median Sell fraction averaged over the final 500 steps for runs without a majority. The dealer panels used a single starting market that reached a majority at dealer depths $h = 0.1$, 1, and 100 for fraction $f = 0.20$, chosen to minimize the summed distance from the median onset among majority-reaching runs at each depth. The same starting market also supplied a comparison with technical traders held at dealer depth $h = 100$. We extended the five examples across fractions to 10,000 steps after the rule switch, with snapshots at 0, 1,500, 3,000, 5,000, and 10,000 steps (Fig. 4). The dealer comparisons retained their 3,000-step duration and snapshots at 0, 500, 1,500, and 3,000 steps. Selection therefore represented typical outcomes within each group over the original observation period, and a market without a majority at that endpoint could reach one later. Individual runs illustrated how groups of sellers developed, while probabilities across runs measured how often a majority occurred.

3 Results

We first asked whether a lasting change in how some traders respond to their neighbors could ~~turn a mixed market into one dominated by sellers~~push a market of mixed Buy, Hold, and Sell decisions to a Sell majority. Rule 19286 differs from the baseline Rule 19280 in a single response: a trader with three neighbors in Buy and one in Hold chooses Sell rather than Buy (Fig. 1A,B). We paired each switched market with a no-switch control that kept Rule 19280 and shared its baseline history (Supplementary Table S1). With active technical traders and dealer depth $h = 100$, none of the 50 starting markets reached a Sell majority within 3,000 steps after the switch when the fraction of interior traders given the new rule was $f = 0.10$ or 0.15~~, whereas all~~. All of them did at $f = 0.20$ and 0.25~~. No~~, whereas no control reached a majority over the same period. The median onset fell from 1,~~070.0 steps~~070 steps after the switch at fraction $f = 0.20$ to 311.5 steps at $f = 0.25$ (Supplementary Table S2). In the displayed pair at fraction $f = 0.20$, groups of sellers expanded after the rule switch while the control kept its mixture of decisions (Fig. 1C). ~~A one-time instruction did not have the sameeffect: setting the~~ To test whether a single burst of Sell intentions could do the same, we instead set the intentions of the same selected traders to Sell once at the switch and let them keep Rule 19280 for every later update. This one-time instruction produced no majority at fraction $f = 0.15$, 0.20, or 0.25~~produced no majority in either technical-trader condition~~, whether or not the technical traders responded to price (Supplementary Table S5). Which response changed also mattered. ~~Of the eighteen alternative rules tested at fraction $f = 0.20$, each differing~~ We screened all eighteen rules that differ from Rule 19280 in a single response, ~~6~~ including Rule 19286, over 20 starting markets with technical traders active and held. At fraction $f = 0.20$, six rules produced a majority in every ~~tested starting~~ market under both ~~technical-trader conditions, while 12 produced none (~~conditions. Five of them placed Sell at one of the neighbor sums 1 to 5, the only sums at which Rule 19280 does not already assign Sell, and Rule 19286 was one of these five. The sixth replaced Buy with Hold at neighbor sum 3. The other twelve produced no majority in any market: ten moved a response away from Sell, and two replaced Buy with Hold at neighbor sums 1 and 2 (Fig. 1B; Supplementary Table S6). Thus, ~~some tested lasting rule switches~~ the lasting switch to Rule 19286 produced majorities where a one-time Sell ~~instructions~~ instruction to the same traders did not, and outcomes depended on which response changed and how many traders received it.

~~Because technical traders respond to the quoted price,~~ The rule screen produced majorities with technical traders held as well as active, which suggested that a Sell majority ~~might have required~~ did not require price feedback. To ~~determine whether it did~~test this across fractions and dealer depths, we repeated the rule switch ~~with every technical trader~~ at 11 fractions between 0.10 and 0.25 with the technical traders either active or held in Hold (Fig. 2). We also varied the dealer depth over $h = 0$, 0.1, ~~so that~~ 0.5, 1, 2, 5, 10, and 100, because depth sets how far the quoted price moves for a given net trade. In the held condition, trades still completed and, at positive depths, the price still moved, but no decision responded to it~~(Fig. 2). With technical traders held ,~~. Because no decision in the held condition depended on cash, shares, or price, the ~~Sell-majority probability was 0.34 at fraction $f = 0.175$, 0.70 at $f = 0.185$, and 1.00 at $f = 0.20$ (Fig. 2A). For each starting market and fraction,~~ onset, final mean Sell fraction, and final interior grid for each starting market and fraction were identical across dealer depths despite different price paths. ~~Without technical-trader responses, financial variables cannot influence decisions, so responses~~ The held Sell-majority probability was therefore the same at every depth: 0.34 at fraction $f = 0.175$, 0.70 at $f = 0.185$, and 1.00 at $f = 0.20$ (Fig. 2A). Responses to neighbors were thus sufficient to produce a Sell majority. ~~Allowing technical traders to respond to price changed majority probability near~~ Majority probability rose from zero to one over a narrow range of fractions, which we refer to as the crossover. Within it, the probability was usually higher with technical traders active than held. At dealer depth $h = 100$, ~~25 of 50 markets reached a majority~~ the probability with technical traders active was 0.50 at fraction $f = 0.175$, compared with ~~17 when held. At fraction $f = 0.185$, the corresponding counts were 42 and 35~~ 0.34 when held, and 0.84 at $f = 0.185$, compared with 0.70 (Fig. 2A~~,B). Both the size and the sign of the difference in majority probability between active and held conditions depended on the dealer depth and fraction~~). Across the 88 combinations of eight depths and 11 fractions, the active minus held difference was positive in 22, with a largest value of 0.18, and negative in two, by at most 0.04. It was zero in the rest, including every fraction outside the crossover and every combination at $h = 0$, where the price never moved, the technical traders' band kept zero width, and they stayed in Hold (Fig. 2B). Thus, price feedback could change how often the rule switch produced a majority, but it was not needed for selling decisions to spread.

We next asked how the spread of selling depended on the number of interacting traders. With technical traders held, we compared markets whose grids had $L = 32$~~to $L = 128$~~, 48, 64, 96, and 128 rows and columns~~and refined the~~ , adding tested fractions near the rise in majority probability at each size ~~, keeping the same starting markets~~ and using the same 50 starting markets at every added fraction (Fig. 3). ~~Larger gridsgave a sharper response. The~~ We repeated the comparison with technical traders active at dealer depths $h = 1$ and 100. In larger grids, majority probability rose more steeply with the fraction f and at smaller fractions. With technical traders held, the fraction at which half the markets reached a majority within 3,000 steps, f_{50}, fell from 0.216 at grid size $L = 32$ to 0.166 at $L = 128$~~, and the width between majority probabilities of~~ . At every size, both active midpoints were within 0.007 of the held value, usually at or below it, and at $L = 128$ they were within 0.001 (Fig. 3C). These midpoints changed little when a logistic curve replaced interpolation: across all size and technical-trader conditions, the two estimates differed by at most 0.0035, less than a tenth of the shift across grid sizes (Supplementary Tables S3 and S10). The width $f_{75} - f_{25}$, the range of fractions over which the held majority probability rose from 0.25 ~~and~~ to 0.75~~narrowed from~~ , narrowed about threefold over the same sizes (Fig. 3A). With 95% intervals from resampling whole starting markets in brackets, it was 0.023 [0.018, 0.027] ~~to~~ at $L = 32$ and 0.008 [0.004, 0.010] ~~, with brackets giving 95% intervals from resampling whole starting markets (Fig. 3A,C;~~ at $L = 128$ (Supplementary Table S3). The susceptibility ~~, which measures how much the~~ multiplies the variance of the final mean Sell fraction ~~varied~~ between starting markets ~~, rose from a sampled maximum of~~ by the number of interior traders, so it could not exceed 0.25 if each trader ended in Sell independently, because the variance of an average over N independent traders is at most $1/(4N)$. With technical traders held, its largest value over the tested fractions was 21.2 at grid size $L = 32$ ~~to~~ and 241.2 at $L = 128$ (Fig. 3B,D). ~~The midpoints changed little when a logistic curve replaced interpolation: across all size and technical-trader conditions, the two estimates differed by at most 0.0035 (Supplementary Tables S3 and S10)~~These values, far above that bound, showed that the traders within each market ended in Sell together rather than independently. Observation time ~~still mattered~~also moved the midpoint. At grid size $L = 64$, extending the held markets threefold, from 3,000 to 9,000 steps~~moved the midpoint~~ , lowered it from 0.178 to 0.167. This shift of 0.011 was similar to the change of 0.012 from $L = 64$ to $L = 128$ at 3,000 steps, so over these ranges the midpoint depended on observation time about as much as on grid size. Longer observation lowered

the midpoint at each size tested this way, $L = 32$, ~~and alternative Sell thresholds and size-dependent horizons gave further comparisons~~ 64, and 128, including when the Sell fraction required to count a run was set to 0.4 or 0.6 instead of 0.5 (Supplementary Table S9). Thus, ~~the size comparison described a crossover within~~ larger grids sharpened the rise in majority probability and moved it to smaller fractions. The position of the rise also moved with observation time, so the size comparison located a crossover for a finite observation period rather than a ~~fixed threshold~~ fraction that would hold for indefinitely long runs, and these runs did not determine whether such a fraction exists.

~~A 3,000-step observation period could end before a slow majority formed, so~~ Because the midpoint moved with observation time, we extended five example markets with active technical traders and dealer depth $h = 100$ to 10,000 steps after the switch, to see whether markets without a majority at 3,000 steps reached one later and how the others developed (Fig. 4). ~~Selection used~~ Before any grid images were inspected, we had selected them from the 3,000-step outcomes ~~, before image inspection. Two had not reached a majority by that endpoint and three had. Of the two~~ as typical of their fraction and outcome. They comprised one market without a majority at fraction $f = 0.15$, one with and one without a majority at $f = 0.175$, and one with a majority at each of $f = 0.185$ and 0.20. The market without a majority ~~, the market at fraction~~ at fraction $f = 0.15$ still had none after 10,000 steps, whereas the market without one at $f = 0.175$ reached ~~one after~~ a majority 4,073 steps ~~, whereas the market at $f = 0.150$ had not reached one by~~ after the switch. That market's maximum drawdown rose from 10.17% at 3,000 steps to 26.34% at 10,000 steps ~~. The three earlier majorities formed after~~ (Fig. 4B; Supplementary Table S11). The other three majorities had formed 2,330, 1,836, and 1,028 steps after the switch at fractions $f = 0.175$, 0.185, and 0.20, respectively (Supplementary Table S11). Each displayed majority formed as groups of sellers grew and merged (Fig. 4A,C). ~~For the market that~~ At fraction $f = 0.15$, no switched market had reached a majority ~~late, maximum drawdown rose from 10.17% at~~ by 3,000 steps~~to 26.34% at 10, 000 steps (Supplementary Table S11)~~, so any majority there would form late. To measure ~~late-majority frequency at fraction $f = 0.150$~~how often this happened, we extended to 10,000 steps the runs from all 50 starting markets ~~and~~ at that fraction, with the same active technical traders and dealer depth, together with their no-switch controls. Two switched markets reached a majority by 10,000 steps, compared with none at 3,000 steps, and no control reached one. ~~Among the switched markets~~The longer runs also showed how the switch affected prices before any majority formed. Every control stayed below a maximum drawdown of 1.15% over 10,000 steps. Among the 50 switched markets at this fraction, the median drawdown ~~changed from~~ was already 5.04% ~~to~~ at 3,000 steps, before any of them had a majority, and 5.87% ~~, whereas the largest drawdown rose from~~ at 10,000 steps. Their largest drawdown was 12.29% ~~to~~ at 3,000 steps. At 10,000 steps it was 25.30%, in a different market, one that first reached a majority between 3,000 and 10,000 steps (Supplementary Table S11). Thus, remaining ~~mixed~~ without a majority for 3,000 steps did not establish a long-run state~~; majority outcomes depended on the observation period~~, the rule switch deepened price declines even before any majority formed, and a majority that formed later could raise a market's drawdown well above its value at 3,000 steps.

We next asked how dealer reserves affected ~~majority probability and the resulting price decline . We varied the dealer depth h, the dealer 's initial shares per trader, while keeping the same rule switch (Fig. 2). Near the crossover, dealer depth also affected majority probability. We tested whether the technical traders' adaptive band widths contributed to that dependence. At fraction $f = 0.175$, the Sell-majority probability was 0.44 at dealer depth $h = 0.1$ and 0.50 at $h = 100$, a difference of -0.06 that became -0.26 when the band widths were frozen~~ the price decline that accompanied a majority, because a dealer with small reserves might be expected to limit a decline by buying fewer of the shares that traders offered. Using the same dealer-depth comparison, we examined the quoted price after the rule switch ~~. We matched starting markets across depths and band conditions. Subtracting the frozen-width contrast from the adaptive-width contrast gave a paired difference of 0.20 with a 95% bootstrap interval of -0.08, 0.50, leaving the contribution of band-width adaptation unresolved (Supplementary Table S7~~(Fig. 2). At fraction $f = 0.20$, every starting market reached a majority at every tested depth, ~~yet with active technical traders~~ with technical traders active or held. The declines nonetheless differed in size: with active technical traders, the median maximum drawdown of switched markets was 99.99% at dealer depth $h = 0.1$, 99.52% at $h = 1$, and 25.15% at $h = 100$ (Fig. 2C). ~~In an additional comparison, median drawdowns were~~ Additional runs at dealer depths $h = 40$ and 400, outside the eight-depth sweep, at the same fraction and with active technical traders, followed the same decrease with depth, with median drawdowns of 48.62% ~~at dealer depth $h = 40$~~ and 7.08% ~~at $h = 400$~~ (Supplementary Table S8). Smaller reserves made the quoted price more sensitive to net selling, because the ~~reserves set both the dealer's available funds and the price~~

~~required to execute a net trade. Much of the decline at dealer depth~~ same net sale added a larger proportion to a small share reserve. Shallow positive reserves did not limit these declines by refusing trades. A constant-product dealer never ran out of cash for buying, and its few rejections came from clearing limits: median dealer-rejection fractions were 226, 34, and 0.6 per million eligible orders at dealer depths $h = 0.1$~~occurred~~ , 1, and 100 (Fig. 2D). The same price sensitivity also produced large declines without the rule switch~~: no-switch controls there already~~ . At shallow depth, ordinary order imbalances moved the quoted price up and down by large amounts, so the largest fall from an earlier peak was large even without a lasting trend. No-switch controls at dealer depth $h = 0.1$ had a median drawdown of 77.04%, compared with 0.75% at $h = 100$. The median ~~paired~~ over starting markets of each switched market's increase in drawdown ~~relative to the~~ over its no-switch control ~~was~~ rose from 22.96 percentage points at dealer depth $h = 0.1$ ~~and~~ to a largest value of 80.31 percentage points at $h = 5$ and fell to 24.39 percentage points at $h = 100$ (Supplementary Table S2). ~~Small positive reserves accompanied these declines while rejecting few orders backed by trader resources, with median dealer-rejection fractions of 0.000226 at dealer depth~~ At $h = 5$, the median drawdown was 14.03% in controls and 94.43% in switched markets. The increase was smaller at both ends because at $h = 0.1$ ~~, 0.000034 at $h = 1$, and 0.0000006 at~~ the controls' decline left little room for more, whereas at $h = 100$ ~~(Fig. 2D).~~ the quoted price responded weakly to net selling. Zero reserves were a separate ~~limit~~case, in which only opposing trader orders matched, the dealer ~~'s net trading was zero~~did not trade, and the price stayed fixed, so the drawdown was zero although every market reached a majority. In the displayed starting market 2 at fraction $f = 0.20$, a majority formed at dealer depths $h = 0.1$, 1, and 100 with active technical traders~~while the price followed distinct paths~~ , while the quoted price fell nearly to zero at the two shallower depths and by about a quarter at $h = 100$ (Fig. 5). Thus, ~~across the tested depths~~ at fraction $f = 0.20$, the dealer's reserves set ~~the severity of the decline rather than whether a majority formed~~how far the quoted price fell both with and without the rule switch, and shallow positive reserves did not prevent severe drawdowns.

Trades also depended on the traders' own resources, so we independently ~~changed~~ multiplied their cash and shares by 0.1, 1, or 10 at the rule switch, at dealer depths $h = 0.1$, 1, and 100 (Fig. 6). ~~With technical traders held, every cash and share combination at~~ We report fraction $f = 0.18$~~produced the same decision history within a starting market, and the Sell-majority probability was 0.64 at every dealer depth~~, near the held midpoint of 0.178 for the 64×64 grid used here, and give three neighboring fractions in Supplementary Table S4. Holdings entered the order and clearing calculations but not the interior decision rule, so they could change decisions only through the price response of active technical traders. With technical traders held, every cash and share combination produced the same decision history within a starting market, and the Sell-majority probability was 0.64 at every dealer depth, with a 95% Wilson interval of [0.50, 0.76]. With technical traders active, the probability was 0.58 to 0.60 with unchanged resources, slightly below the held value, and ranged from 0.54 to 0.80 across all resource combinations and dealer depths (Fig. 6A~~; Supplementary Table~~ S4). This comparison had no separate no-switch controls, so its drawdowns describe switched markets rather than effects of the switch. Among the 27 active combinations, the five highest majority probabilities all had the share multiplier 0.1, and within each combination of cash and dealer depth that multiplier also gave the smallest drawdown, so a higher majority probability did not bring a larger drawdown (Fig. 6D). With unchanged cash and dealer depth $h = 100$, share multipliers of 0.1, 1, and 10 gave median drawdowns of 3.17%, 20.18%, and 77.55% (Fig. 6B)~~, because more sharessupplied more potential sales while the reserve curve set the price at which net selling could clear~~. Drawdowns rose with shares because fewer Sell orders were blocked for lack of shares, so more net selling reached the dealer's reserves, which the intervention did not rescale. By contrast, raising the cash multiplier from 1 to 10 at dealer depth $h = 1$ with unchanged shares changed the median drawdown only from 96.46% to 97.10% and the majority probability from 0.58 to 0.56, in markets where the decline was already close to complete (Fig. 6A,B). ~~Few dealer rejections did not imply a small drawdown across these resource combinations~~ Dealer rejection did not cushion these declines: median dealer-rejection fractions stayed below 5,000 per million eligible orders in every combination, including those with the largest drawdowns (Fig. 6~~C,D~~B,C). Together, ~~these~~ the dealer-depth and resource comparisons showed that financial outcomes in switched markets depended on traders' cash and shares and on the dealer's reserves~~. A majority decision to sell~~ , so a Sell majority did not by itself determine ~~how many shares could change hands or~~ how far the quoted price would fall.

Finally, we asked whether trading costs changed the spread of selling or its financial consequences, repeating the paired rule switch with fees charged on completed orders. Because fees did not enter the interior decision rule, the Sell fraction at every step and the final grid were identical across all four fee rates and three dealer depths when technical traders were held, for each starting market and fraction. With technical traders active and fraction $f = 0.20$, the Sell-majority probability across fee rates was 1.00 at dealer depth $h = 0.1$, 0.98–1.00 at $h = 1$, and 1.00 at $h = 100$ (Supplementary Table S12). Nearer the crossover, at fraction $f = 0.175$ and dealer depth $h = 100$, the probability changed from 0.50 without a fee to 0.40 at 25 basis points. The paired difference was -0.10 with a 95% interval of [-0.28, 0.08], so its direction was not resolved. The effect of fees on price declines depended on the condition. At dealer depth $h = 100$ and fraction $f = 0.20$, a 25-basis-point fee lowered the median maximum drawdown from 25.15% to 23.67%, a median paired change of -1.34 percentage points with a 95% interval of [-1.60, -1.16]. At dealer depth $h = 1$ and fraction $f = 0.175$, the same fee raised it from 76.52% to 92.81%, a median paired change of 9.04 percentage points with a 95% interval of [3.46, 16.98]. Over a complete run, the fees accumulated: at 25 basis points, dealer depth $h = 100$, and fraction $f = 0.20$, the median cumulative fee income of the dealer was 36.19% of the traders' initial cash (Supplementary Table S12). Thus, fees did not reduce price declines in every tested condition, and because the decision rules were unchanged, any effect of fees on interior decisions required active technical traders responding to the price.

4 Discussion

In this study, we explored how a lasting change in the way a minority of traders formed their intentions could spread selling through a market, and what then determined how far the price fell. A Sell majority formed once enough traders received the new response, even though every other trader kept its original rule (Fig. 1), whereas the size of the resulting price decline also depended on trader resources and the dealer's reserve curve. The changed response acted whenever its neighbor configuration recurred, so the selected traders did not need to remain in Sell for the change to keep influencing the market. The tested rule switch produced majorities that a one-time instruction to sell did not (Supplementary Table S5). Testing alternative single-response changes showed that majority formation depended on which response changed, limiting the conclusions to the tested rules (Supplementary Table S6). Because each switched market shared its history with a no-switch control, the difference between the two measured the effect of the switch at that starting market, and repeating the pair across starting markets measured how often a majority formed within the observation period [30, 31]. These conclusions concern assigned rule changes and the spread of selling intentions: traders did not learn the new rule, adopt it for its profitability, or pass the rule itself to other traders.

The spread of selling did not require any decision to respond to the price. With the technical traders kept in Hold, trades still completed and the price still moved, but no decision responded to it, and Sell majorities still formed (Fig. 2). In the displayed markets with active technical traders, a majority formed as groups of sellers grew and merged across the grid (Fig. 4). These patterns resemble the growth of ordered regions in other interacting systems away from equilibrium [32, 33], and the dependence on the fraction of traders given the new rule connects the model to accounts of how local changes spread through a population [34–37]. Larger grids sharpened the rise in majority probability, but the switched fraction at which half the markets reached a majority changed with observation time (Fig. 3; Supplementary Table S9), so a market that remained mixed at one endpoint could still reach a majority later. Neither the displayed markets nor the sampled majority-probability curves established a threshold for indefinitely large or long-lived markets [27, 28]. The resemblance to droplets forming and joining also did not establish classical nucleation, because we measured no energy barrier, surface tension, formation rate, or critical group size separating growth from decay. Thus, the supported mechanism was the spread of selling through neighboring decisions on finite grids over finite observation periods.

How far the price fell depended on how the dealer's resources entered the price. Dealer models commonly connect trading to intermediary cash and inventory [38–41], but holdings alone do not specify how those resources translate into prices. Here, the constant-product reserve curve set both the uniform price at which a batch of orders executed and the quoted price left after settlement [23]. When the dealer bought shares, its share reserve rose and its cash reserve fell, so continued net selling lowered execution prices. Shallow positive reserves therefore allowed large proportional declines

while rejecting few eligible orders (Fig. 2C,D). Zero dealer depth was a separate case, in which the dealer took no net position, only opposing trader orders matched, and the price stayed fixed, so it could not be read as evidence that every small dealer protects the market from a crash. The no-switch controls mattered as well, because markets with shallow reserves had large drawdowns even without the rule switch, and only the paired increases separated that baseline decline from the effect of the switch (Supplementary Table S2). Thus, the size of a crash depended on the execution and price mechanism as well as on the resources available, and scarce resources did not by themselves limit the decline.

Financial conditions could still change later decisions, but only through the technical traders, who responded to the quoted price. Changing dealer depth or trader holdings altered completed orders and prices, which changed the technical traders' decisions and therefore the neighboring intentions seen by adjacent interior traders (Figs. 2 and 6). With the technical traders held, the same financial changes left every interior decision unchanged, which made the held condition an exact control for this effect. With technical traders active, the majority probability neither rose nor fell consistently with dealer depth, even though drawdown at switched fraction $f = 0.20$ fell as positive depth increased. Technical traders' Buy and Sell thresholds lay around a moving average price, at distances that normally adapted over time. Fixing those distances at each market's pre-switch values preserved price feedback, but the paired comparison with adapting distances was too uncertain to establish whether adaptation explained majority-probability differences across dealer depths (Supplementary Table S7). Varying dealer depth changed funding and initial price sensitivity together, so the depth comparisons could not separate the two effects (Supplementary Table S8). Drawdown also depended on our choice of a reserve mechanism; other models link order flow to prices through dealer inference or order-book dynamics [42–44]. Majority probability measured the share of markets reaching a Sell majority, whereas drawdown measured the largest price decline. Neither outcome alone determined the other.

The comparison with established agent-based markets concerns how traders interact and how their rules change: we assigned a persistent rule switch instead of modeling learning or strategic choice. In the minority game, traders choose among strategies scored against shared past outcomes and interact through the market outcome [45, 46]. In the Santa Fe artificial stock market, traders learn forecasting rules from a shared price and dividend [5]. Related studies of the Santa Fe market examined incentives to adopt technical rules [47] and choose how often to revise forecasts [48]. Models with chartists and fundamentalists let traders move between the two groups at rates set by recent profits or by others' opinions [7, 9, 49]. Lattice and cellular-automaton markets encoded local interactions through stochastic imitation or spin-like responses, sometimes alongside market-wide signals [10, 11, 17–20, 50]. Wolfram's one-dimensional market and a later model built on Rule 110 used deterministic rules and set the price from the numbers of buyers and sellers [14, 15]. Here rules did not evolve through learning: the intervention changed one response for a chosen fraction of traders, and the paired control retained the baseline rule. Only technical traders along the edge responded to price, allowing the rule switch to be tested with price feedback active or removed. Orders needed cash or shares and were completed by other traders or by a dealer with limited holdings, so a Sell majority and a price crash were measured separately. These choices gave up the learning and strategic behavior of established models in exchange for a rule space that can be enumerated and an intervention that can be repeated exactly. The questions answered were correspondingly narrower: not whether technical trading emerges or whether the market becomes efficient, but how many traders had to change a rule before a Sell majority formed, whether its formation required feedback from price, and how resources and execution determined the resulting price decline.

Several modeling choices limit how far these conclusions extend. Traders followed assigned rules rather than choosing strategies to maximize returns, and they updated together on a regular grid under one baseline lookup table in which different neighbor configurations with the same sum produced the same response. A trader that had sold all its shares could still influence its neighbors and count toward a Sell majority, because we assumed that intentions remained observable and influential even when a trader could not act on them. We did not test how the results would change if a trader's resources also limited the intentions its neighbors could see. The technical traders followed one band rule and occupied selected positions along the edge of the grid, so other placements and price-based strategies remain untested. Unit orders cleared at a common base price against a constant-product dealer, a specific reserve mechanism rather than a calibrated representation of an equity exchange. The dealer collected prescribed fees in the fee comparisons but did not set its charges or its participation to

maximize profit, and we did not model incentives to provide or withdraw liquidity. The model had no order book, dividends, borrowing, or strategic risk management by the dealer, and its steps and price response were not fitted to market data, so its simulated crashes did not predict the timing or magnitude of actual market crashes. Settlement conserved cash and shares, and an isolated round trip restoring dealer shares also restored dealer trading cash. Neither property established economic realism or ruled out profits from other traders. The long no-switch controls extended observation of existing starting markets without establishing their behavior indefinitely (Appendix A.5), and neither the baseline period nor an average over final states showed that the distribution of market states had stopped changing. Other decision rules, asynchronous updates, alternative interaction networks, and different execution mechanisms would test how the connection between local decisions and price declines depends on these assumptions.

5 Conclusion

A lasting change in the way a minority of traders formed their intentions could turn a mixed market into one dominated by sellers, even when no trader's decision responded to the price. The fraction of traders needed depended on the number of interacting traders and on how long the market was observed, and active technical traders let financial conditions change later decisions. A Sell majority did not by itself determine the price decline. Trader cash and shares limited which orders could be completed, and the dealer's reserve curve set the prices at which net trades could clear. Shallow positive reserves accompanied severe declines, whereas zero dealer depth allowed only trades between traders at a fixed price. Limited resources therefore did not protect the market from a simulated crash in general. The financial consequence of widespread selling depended on how resources entered execution and price formation, not only on how many traders decided to sell.

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Tables

Table S1: Design of the four main campaigns. Active technical traders responded to the quoted price; held technical traders stayed in Hold. All dealers had finite reserves. The size campaign used active technical traders at dealer depth $h = 1$ and 100, and held technical traders at $h = 100$. Counts include newly simulated no-switch controls, with one such control shared across fractions within a starting market and condition where the design permits.

Campaign	Grid sizes	Dealer h	Technical traders	Starting markets	Simulations
Rule switch	64	100	active	50	400
Dealer depth	64	0–100	active/held	50	9,600
Grid size	32–128	$1, 100$	active/held	50	9,750
Cash and shares	64	$0.1, 1, 100$	active/held	50	10,800

Table S2: Paired rule switches on a 64×64 grid over 3,000 post-switch steps. In A, majority probability p counts paths reaching an interior Sell fraction of at least one-half; median onset is conditional on reaching that value. Undefined means no run reached a majority. In B, DD is median maximum drawdown in percent. ΔDD is the median paired difference, in percentage points, with a pointwise 95% whole-market bootstrap confidence interval (CI). These contrasts separate the rule switch from baseline financial variation.

A. Rule switch at $h = 100$, active technical traders

Fraction f	Starting markets	No-switch p	Switched p	Median onset
0.10	50	0.00	0.00	Undefined
0.15	50	0.00	0.00	Undefined
0.20	50	0.00	1.00	1070.0
0.25	50	0.00	1.00	311.5

B. Dealer depth at $f = 0.20$, active technical traders

h	Starting markets	No-switch DD	Switched DD	Paired ΔDD [95% CI]
0	50	0.00	0.00	0.00 [0.00, 0.00]
0.1	50	77.04	99.99	22.96 [22.56, 23.74]
0.5	50	42.83	99.82	57.03 [56.25, 57.86]
1	50	32.92	99.52	66.55 [65.56, 67.71]
2	50	23.15	98.59	75.29 [74.78, 75.89]
5	50	14.03	94.43	80.31 [79.22, 80.96]
10	50	8.31	85.68	77.33 [76.71, 77.48]
100	50	0.75	25.15	24.39 [24.20, 24.47]

Table S3: Grid-size comparisons over 50 starting markets and 3,000 post-switch steps. Brackets give 95% confidence intervals (CI) from 2,000 whole-market bootstrap samples. n_{int} counts fractions with $0.05 < \hat{p} < 0.95$. Crossings use first upward linear interpolation. Susceptibility maxima are selected over the sampled grid, including reselection in each bootstrap sample. Undefined denotes an unbracketed crossing; intervals use available bracketed samples.

Technical traders	h	L	n_{int}	f_{50} [95% CI]	Width [95% CI]	Peak χ_L [95% CI]
Active	1	32	9	0.212 [0.208, 0.217]	0.020 [0.014, 0.028]	26.3 [22.2, 34.4]
Active	1	48	6	0.186 [0.184, 0.188]	0.011 [0.008, 0.019]	54.6 [41.3, 69.9]
Active	1	64	9	0.177 [0.175, 0.180]	0.008 [0.007, 0.013]	85.9 [79.9, 114.6]
Active	1	96	5	0.168 [0.167, 0.169]	0.005 [0.004, 0.007]	143.2 [127.4, 202.5]
Active	1	128	7	0.167 [0.165, 0.168]	0.007 [0.005, 0.009]	219.8 [198.5, 300.9]
Active	100	32	6	0.214 [0.210, 0.218]	0.020 [0.012, 0.028]	25.7 [22.2, 33.9]
Active	100	48	5	0.186 [0.184, 0.188]	0.011 [0.008, 0.022]	51.2 [38.6, 65.3]
Active	100	64	5	0.175 [0.174, 0.180]	0.011 [0.009, 0.014]	88.5 [72.2, 124.2]
Active	100	96	5	0.169 [0.165, 0.171]	0.009 [0.007, 0.011]	132.5 [119.7, 178.7]
Active	100	128	6	0.166 [0.165, 0.168]	0.007 [0.005, 0.009]	230.0 [199.8, 302.8]
Held	100	32	5	0.216 [0.211, 0.224]	0.023 [0.018, 0.027]	21.2 [16.5, 27.2]
Held	100	48	5	0.193 [0.184, 0.195]	0.019 [0.016, 0.022]	48.6 [40.3, 65.3]
Held	100	64	5	0.178 [0.176, 0.180]	0.015 [0.008, 0.019]	110.5 [78.5, 138.5]
Held	100	96	5	0.168 [0.167, 0.170]	0.009 [0.006, 0.012]	168.7 [137.6, 221.0]
Held	100	128	7	0.166 [0.165, 0.167]	0.008 [0.004, 0.010]	241.2 [202.6, 326.0]

Table S4: Probability of a Sell majority after changes in trader cash and shares over 50 starting markets and 3,000 post-switch steps. Each fraction includes 27 combinations of cash, shares, and dealer depth. Held decisions gave the same probability across those combinations; active decisions gave the stated range.

Fraction f	Technical traders held	Technical traders active (range)
0.170	0.22	0.12–0.36
0.175	0.34	0.30–0.58
0.180	0.64	0.54–0.80
0.185	0.70	0.72–0.94

Table S5: A one-time instruction to sell versus a persistent rule switch at the same selected interior traders. Both conditions used dealer depth $h = 100$, 3,000 post-intervention steps, and no-switch controls. Entries are majority probabilities with 95% Wilson intervals.

Technical traders	f	Starting markets	Sell pulse	Persistent rule
Active	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Active	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Active	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Held	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]

Table S6: All single-response changes to Rule 19280 with dealer depth $h = 100$. Each entry gives majority probability with technical traders held / active over 20 starting markets and 3,000 post-switch steps. Outputs are Buy $= 0$, Hold $= 1$, and Sell $= 2$. This descriptive screen used no multiplicity adjustment.

Sum c	Output change	Rule	$f = 0.15$	$f = 0.20$	$f = 0.25$
0	$2 \to 0$	19278	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
0	$2 \to 1$	19279	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 1$	19283	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 2$	19286	0.00 / 0.00	1.00 / 1.00	1.00 / 1.00
2	$0 \to 1$	19289	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
2	$0 \to 2$	19298	0.95 / 0.95	1.00 / 1.00	1.00 / 1.00
3	$0 \to 1$	19307	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
3	$0 \to 2$	19334	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
4	$1 \to 0$	19199	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
4	$1 \to 2$	19361	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
5	$1 \to 0$	19037	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
5	$1 \to 2$	19523	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
6	$2 \to 0$	17822	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
6	$2 \to 1$	18551	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 0$	14906	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 1$	17093	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 0$	6158	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 1$	12719	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00

Table S7: Adjusting versus fixed Hold-band widths at fraction $f = 0.175$ over 50 starting markets. Brackets in the rows give 95% Wilson confidence intervals (CI). Sell is the mean fraction of technical traders in Sell over post-switch steps and starting markets. The moving center continued to update. The probability at dealer depth $h = 0.1$ minus that at $h = 100$ was -0.06 with adaptive widths and -0.26 with frozen widths. Their paired difference was 0.20, with a 95% interval [-0.08, 0.50] from 2,000 whole-market bootstrap samples.

h	Adaptive p [95% CI]	Frozen p [95% CI]	Adaptive Sell	Frozen Sell
0	0.34 [0.22, 0.48]	0.34 [0.22, 0.48]	0.000	0.000
0.1	0.44 [0.31, 0.58]	0.22 [0.13, 0.35]	0.139	0.044
0.5	0.36 [0.24, 0.50]	0.38 [0.26, 0.52]	0.123	0.081
1	0.36 [0.24, 0.50]	0.40 [0.28, 0.54]	0.128	0.086
2	0.44 [0.31, 0.58]	0.40 [0.28, 0.54]	0.117	0.094
5	0.38 [0.26, 0.52]	0.36 [0.24, 0.50]	0.116	0.140
10	0.40 [0.28, 0.54]	0.38 [0.26, 0.52]	0.125	0.163
100	0.50 [0.37, 0.63]	0.48 [0.35, 0.61]	0.103	0.168

Table S8: Additional dealer-depth comparisons at fraction $f = 0.20$ with active technical traders and 50 starting markets over 3,000 post-switch steps. The initial log-price slope per normalized net purchase is approximately $2/h$; this is implied by the reserve curve, not a separate price-impact coefficient. Depth changes funding and price sensitivity together. Drawdown is a percentage, and the dealer-rejection fraction uses post-switch eligible orders. Probability brackets are 95% Wilson intervals.

h	Initial slope $2/h$	Majority probability	Median drawdown (%)	Rejection
40	0.050	1.00 [0.93, 1.00]	48.62	1.41×10^{-6}
100	0.020	1.00 [0.93, 1.00]	25.15	6.04×10^{-7}
400	0.005	1.00 [0.93, 1.00]	7.08	2.82×10^{-7}

Table S9: Observation time and the required Sell fraction with technical traders held and dealer depth $h = 100$. Entries are interpolated f_{50} with 95% whole-market bootstrap intervals over 50 starting markets. The fraction grid is independent of adaptive refinement. Undefined denotes an unbracketed crossing; intervals use available bracketed samples.

L	Post-switch steps	Criterion 0.4	Criterion 0.5	Criterion 0.6
32	750	0.236 [0.228, 0.238]	0.249 [0.245, 0.255]	0.266 [0.260, 0.270]
32	1,500	0.216 [0.210, 0.220]	0.230 [0.228, 0.234]	0.245 [0.238, 0.250]
32	3,000	0.207 [0.205, 0.210]	0.219 [0.210, 0.225]	0.229 [0.226, 0.232]
64	3,000	0.172 [0.169, 0.176]	0.178 [0.176, 0.180]	0.186 [0.180, 0.188]
64	9,000	0.165 [0.163, 0.167]	0.167 [0.165, 0.168]	0.171 [0.169, 0.175]
128	3,000	0.163 [0.162, 0.164]	0.167 [0.165, 0.168]	0.169 [0.168, 0.171]
128	6,000	0.154 [0.153, 0.156]	0.157 [0.156, 0.158]	0.159 [0.158, 0.160]
128	12,000	0.150 [0.148, 0.151]	0.152 [0.150, 0.153]	0.153 [0.152, 0.155]

Table S10: Logistic fits and sampled susceptibility maxima over 50 starting markets and 3,000 post-switch steps. Parentheses give whole-market bootstrap standard errors (SE). The fitted curve imposes increasing probability; empirical crossings do not. The final column gives a sampled fraction, whose precision is limited by the local fraction grid.

Technical traders	h	L	Logistic f_{50} (SE)	Logistic width (SE)	f at peak
Active	1	32	0.212 (0.0015)	0.020 (0.0017)	0.217
Active	1	48	0.185 (0.0013)	0.014 (0.0012)	0.190
Active	1	64	0.177 (0.0006)	0.010 (0.0007)	0.185
Active	1	96	0.168 (0.0005)	0.006 (0.0006)	0.175
Active	1	128	0.166 (0.0005)	0.006 (0.0005)	0.168
Active	100	32	0.215 (0.0015)	0.018 (0.0017)	0.217
Active	100	48	0.185 (0.0011)	0.013 (0.0014)	0.190
Active	100	64	0.177 (0.0008)	0.009 (0.0007)	0.190
Active	100	96	0.168 (0.0006)	0.007 (0.0005)	0.175
Active	100	128	0.166 (0.0004)	0.005 (0.0004)	0.168
Held	100	32	0.218 (0.0016)	0.018 (0.0014)	0.225
Held	100	48	0.189 (0.0011)	0.013 (0.0009)	0.190
Held	100	64	0.178 (0.0010)	0.011 (0.0008)	0.185
Held	100	96	0.169 (0.0006)	0.007 (0.0006)	0.172
Held	100	128	0.166 (0.0005)	0.006 (0.0004)	0.168

Table S11: Longer observation with dealer depth $h = 100$ and active technical traders. Times are steps after the rule switch, following 1,000 baseline steps. Drawdowns and price changes are percentages; price changes use the quoted price at the rule switch as reference. Final Sell averages the last 500 steps of the longer observation period. None means no Sell majority by 10,000 steps. Selection used newly computed short-run summaries before inspection of images. The extensions follow the same starting markets, and are not independent replications.

A. Five examples selected from new 3,000-step results

f	Starting market	First majority	Drawdown 3,000 steps	Drawdown 10,000 steps	Price change 10,000 steps	Final Sell
0.150	2	None	5.22	5.22	-3.29	0.268
0.175	15	4,073	10.17	26.34	-26.32	0.897
0.175	41	2,330	22.14	25.51	-25.49	0.594
0.185	5	1,836	23.77	26.07	-26.03	0.910
0.200	38	1,028	25.61	26.50	-26.48	0.913

B. All 50 starting markets at $f = 0.150$

Decision rule	Steps	Majorities / 50	Median drawdown	Worst drawdown	Median price change
New rule at 15%	3,000	0/50	5.04	12.29	-3.40
New rule at 15%	5,000	0/50	5.25	17.65	-3.49
New rule at 15%	10,000	2/50	5.87	25.30	-3.54
Original rule	3,000	0/50	0.75	1.15	0.06
Original rule	5,000	0/50	0.81	1.15	0.04
Original rule	10,000	0/50	0.88	1.15	0.04

Table S12: Transaction-fee comparisons. Each condition used 50 starting markets. The fee was charged to each completed buyer and seller from initialization. Sell-majority probabilities are followed by Wilson 95% intervals. In Panel A, drawdown and cumulative dealer fees are medians for the switched runs at fraction $f = 0.20$. Drawdown uses the 3,000-step post-switch observation period, including the quoted price immediately before the switch. Fees include all 4,000 trading steps and are expressed as a percentage of initial aggregate trader cash. Fee income was held separately from trading reserves. Panel B reports paired differences between 25 and zero basis points with percentile 95% intervals from 2,000 whole-market bootstrap samples. The probability difference is a difference in proportions; drawdown differences are median paired changes in percentage points. The complete protocol, all six tested fractions, and paired differences are archived with the transaction-fee study.

A. Outcomes at each fee rate

h	Technical	Fee (bp)	Majority, $f = 0.175$	Majority, $f = 0.20$	Drawdown (%)	Fees (%)
0.1	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	0.00
0.1	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	14.19
0.1	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.99	26.25
0.1	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.98	52.85
0.1	active	0	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.99	0.00
0.1	active	5	0.38 [0.26, 0.52]	1.00 [0.93, 1.00]	99.99	13.48
0.1	active	10	0.30 [0.19, 0.44]	1.00 [0.93, 1.00]	99.98	25.73
0.1	active	25	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.98	52.90
1	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.53	0.00
1	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.49	13.01
1	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.45	24.32
1	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	99.32	50.30
1	active	0	0.36 [0.24, 0.50]	1.00 [0.93, 1.00]	99.52	0.00
1	active	5	0.44 [0.31, 0.58]	1.00 [0.93, 1.00]	99.50	12.19
1	active	10	0.46 [0.33, 0.60]	0.98 [0.90, 1.00]	99.48	23.83
1	active	25	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	99.40	49.26
100	held	0	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.66	0.00
100	held	5	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.27	7.12
100	held	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	23.80	14.15
100	held	25	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	23.21	34.91
100	active	0	0.50 [0.37, 0.63]	1.00 [0.93, 1.00]	25.15	0.00
100	active	5	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	24.94	7.29
100	active	10	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	24.51	14.45
100	active	25	0.40 [0.28, 0.54]	1.00 [0.93, 1.00]	23.67	36.19

B. Selected paired comparisons with technical traders active

h	f	Outcome	0 bp	25 bp	Paired difference [95% CI]
100	0.175	Majority probability	0.50	0.40	-0.10 [-0.28, 0.08]
100	0.2	Drawdown (%)	25.15	23.67	-1.34 [-1.60, -1.16]
1	0.175	Drawdown (%)	76.52	92.81	9.04 [3.46, 16.98]

Figures

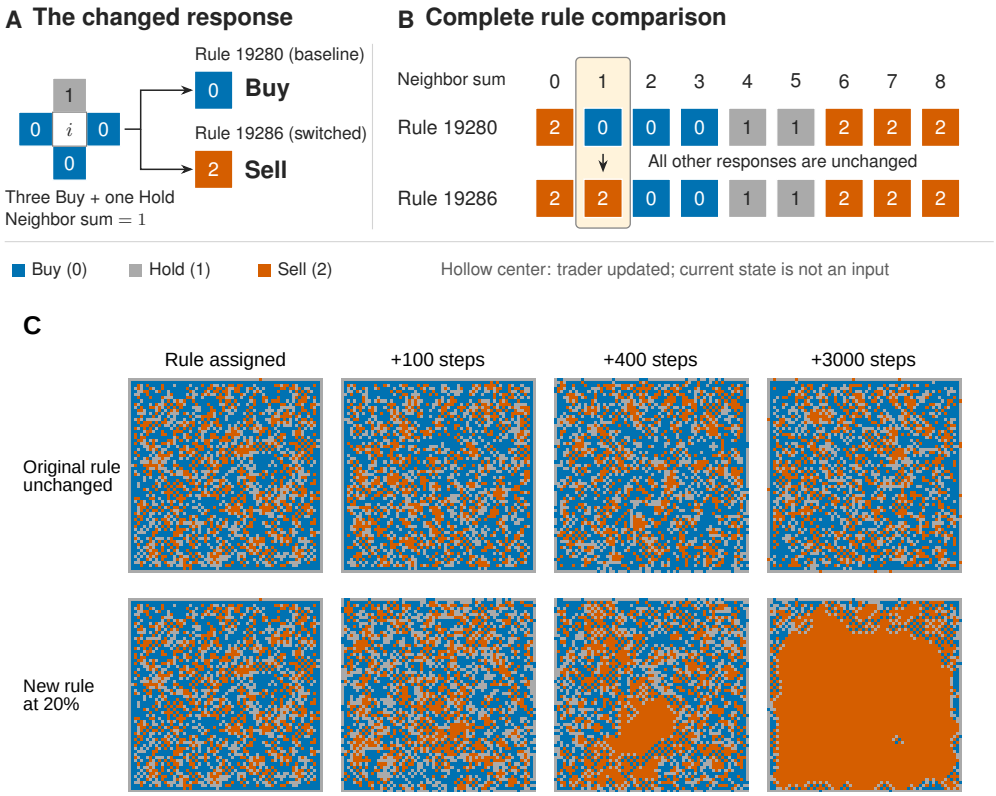


Figure 1: The changed response and paired market decisions. **(A)** The same four neighboring intentions produce different responses at the trader marked i: three Buys and one Hold give neighbor sum 1, producing Buy under Rule 19280 and Sell under Rule 19286. The hollow center's current intention is not an input. Every arrangement with the same neighbor sum produces the same response. **(B)** Complete lookup tables, with the baseline and switched rules aligned by neighbor sum. The highlighted column is the only response changed by the switch. Blue, gray, and orange denote Buy (0), Hold (1), and Sell (2); arrows indicate rule evaluation or a changed response, not completed trades. **(C)** Paired 64×64 grids at the rule switch and after 100, 400, and 3,000 steps. The upper row retains the baseline rule; the lower assigns the new rule to 20% of interior traders. Both paths share a 1,000-step baseline, dealer depth $h = 100$, and active technical traders. Starting market 14 was selected from 50 by proximity to the median paired difference in peak Sell fraction. Ensemble outcomes and conditional onset times appear in Supplementary Table S2.

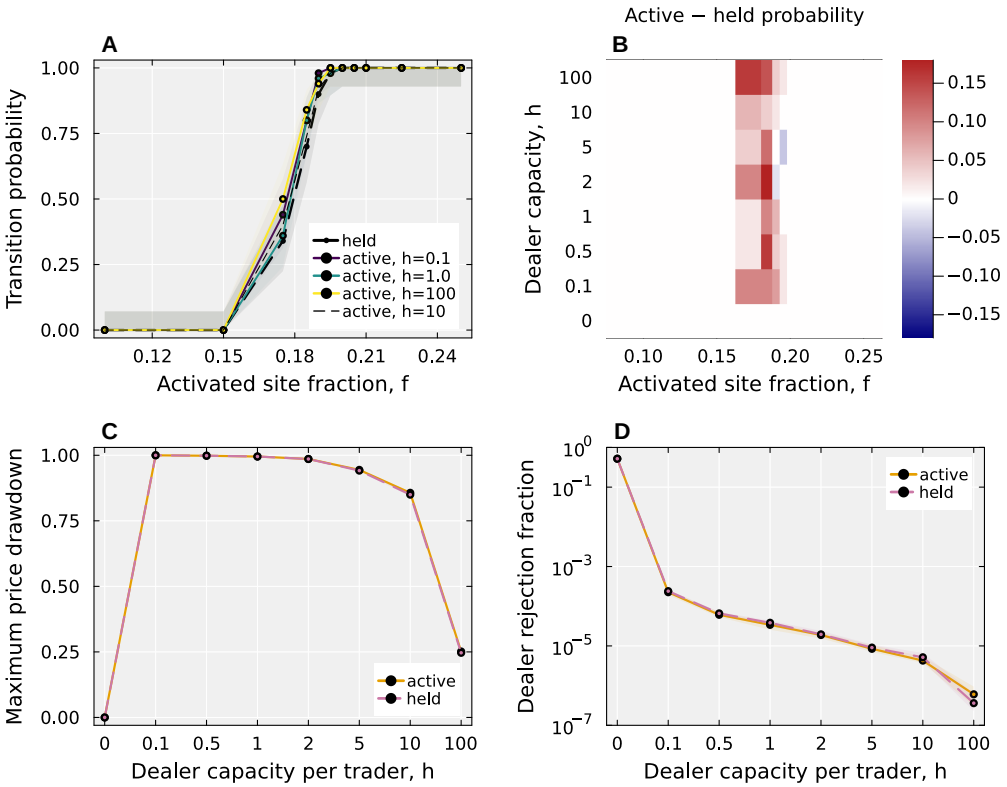


Figure 2: Rule switches across dealer depths over 3,000 post-switch steps. **(A)** Probability of an interior Sell majority with technical traders held or active. Bands are 95% Wilson intervals over 50 starting markets. **(B)** Active-minus-held majority probability at each depth and fraction. **(C)** Median maximum drawdown at fraction $f = 0.20$, with interquartile bands. **(D)** Median fraction of post-switch eligible orders rejected at fraction $f = 0.20$, on a logarithmic axis with interquartile bands. Dealer depth h is initial dealer shares per trader and also determines price sensitivity through the reserve curve. The dashed overlay in A at dealer depth $h = 10$ is a distinct finite-depth comparison; the deep arm is dealer depth $h = 100$. At zero dealer depth only opposing trader orders match and price stays fixed. Panels C and D describe switched markets. No-switch control drawdowns and paired differences appear in Supplementary Table S2.

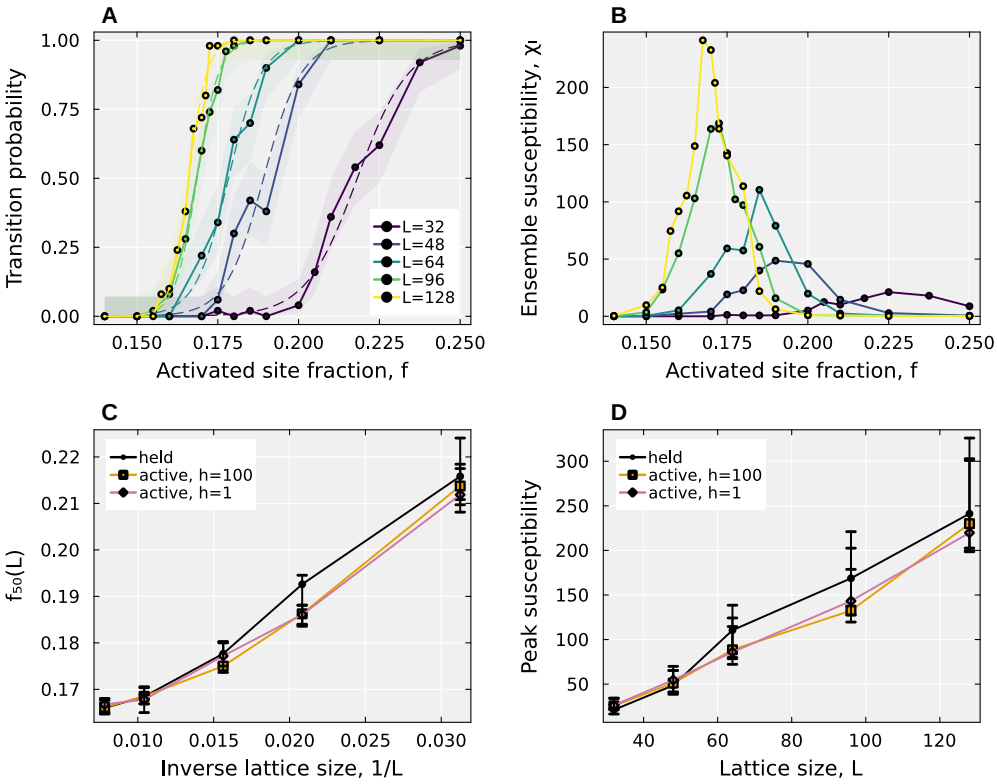


Figure 3: Market size and the formation of a Sell majority over 50 starting markets and 3,000 post-switch steps. **(A)** Majority probability with technical traders held, with 95% Wilson bands. **(B)** Susceptibility with technical traders held: the variance across markets of the final 500-step mean Sell fraction, multiplied by the number of interior traders. **(C)** Interpolated fraction $f_{50}(L)$ at which half the markets reached a majority, against inverse grid size, with technical traders held at dealer depth $h = 100$ and active at $h = 1$ and $h = 100$. **(D)** Sampled maximum susceptibility versus grid size. Error bars in C and D are 95% whole-market bootstrap intervals. Solid lines connect tested values; dashed curves in A are logistic fits. No law for indefinitely large grids was fitted.

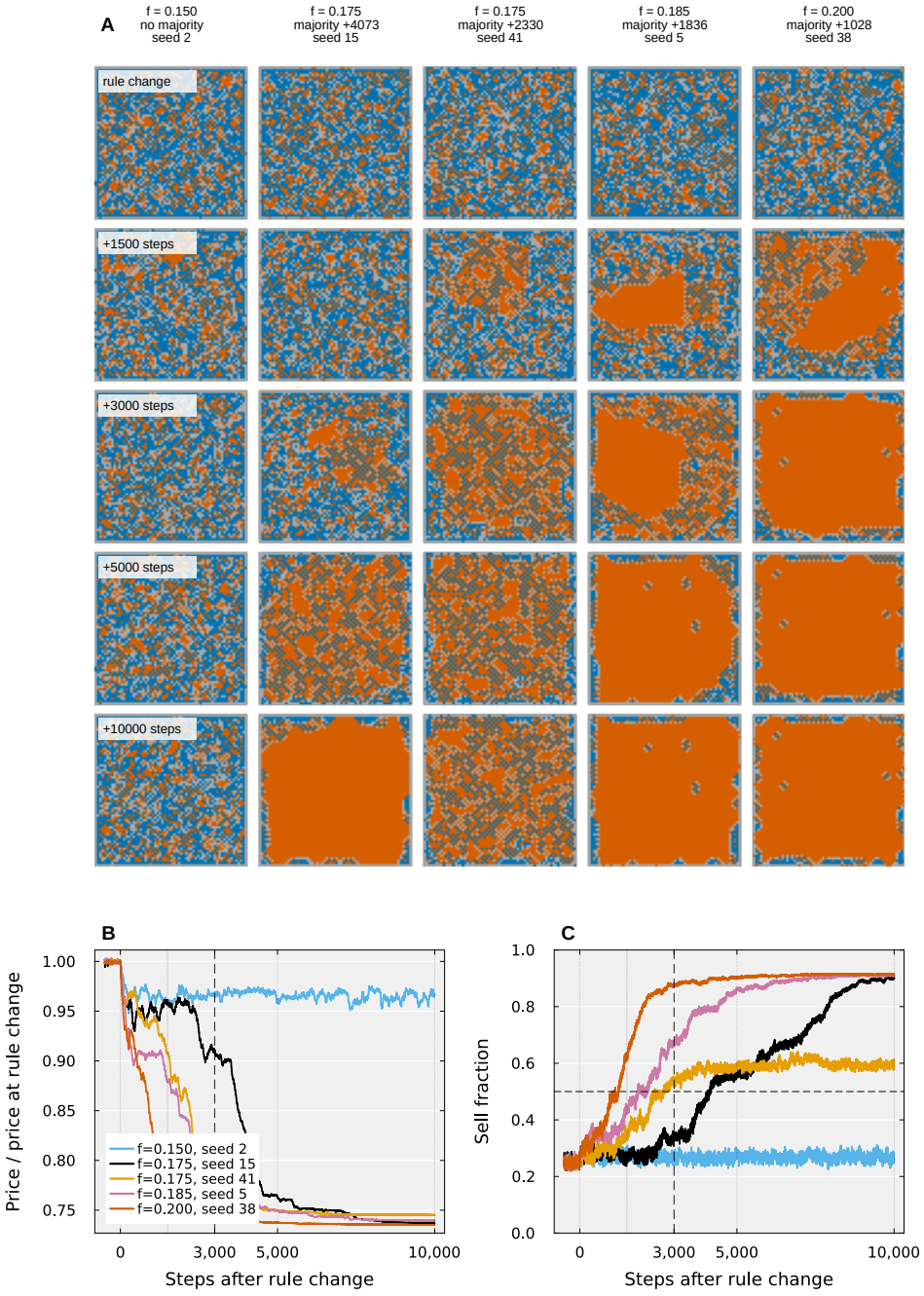


Figure 4: Longer histories of five numerically selected markets with active technical traders and dealer depth $h = 100$. Blue, gray, and orange denote Buy, Hold, and Sell. **(A)** Grid decisions at the rule switch and after 1,500, 3,000, 5,000, and 10,000 steps. Selection used the newly computed 3,000-step summaries before inspection of images. At that endpoint, the first two markets had not reached a majority and the other three had. Nonmajority examples were nearest their group's median final Sell fraction; majority examples were nearest their group's median onset, with the starting-market number breaking ties. Starting markets were 2, 15, 41, 5, and 38, from left to right. The longer outcomes did not affect selection. **(B)** Quoted price divided by its value at the rule switch. **(C)** Interior Sell fraction, with one-half marked by a horizontal line. The dark vertical dashed line marks the original 3,000-step endpoint; other guides mark the remaining snapshots. Onsets and financial summaries appear in Supplementary Table S11.

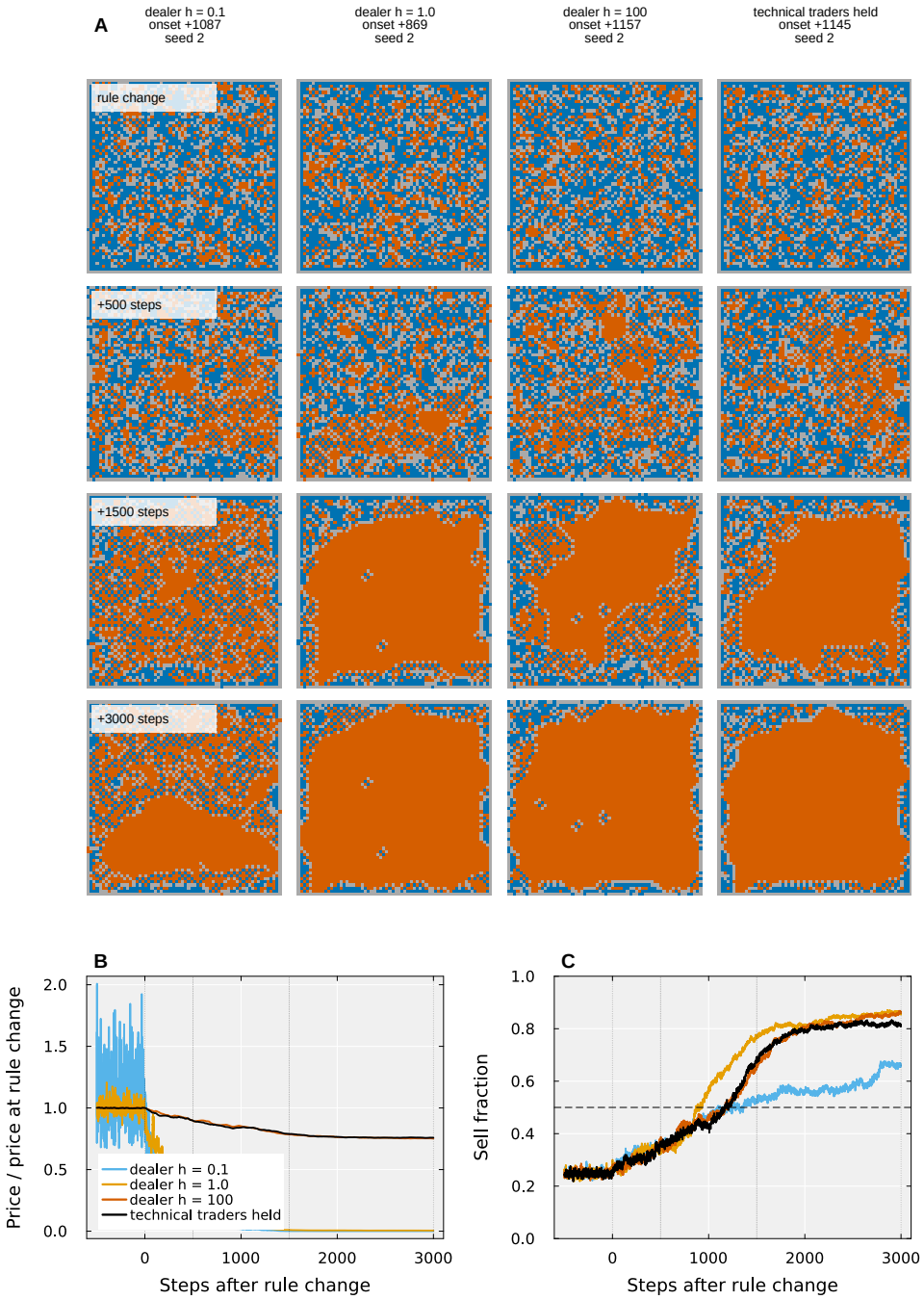


Figure 5: The same starting market at different dealer depths with fraction $f = 0.20$. **(A)** Grid decisions at the rule switch and after 500, 1,500, and 3,000 steps. The first three columns have active technical traders at dealer depth $h = 0.1$, 1, and 100; the last holds the technical traders at $h = 100$. Starting market 2 was selected among markets reaching a majority at all three dealer depths with active technical traders by minimizing the summed distance from their depth-specific median onsets, with the starting-market number breaking ties. **(B)** Quoted price relative to the rule switch. **(C)** Interior Sell fraction. The held condition prevents price feedback into decisions while retaining full trading and price calculations. The selected paths illustrate the same initialization across conditions; ensemble probabilities and drawdowns appear in Fig. 2.

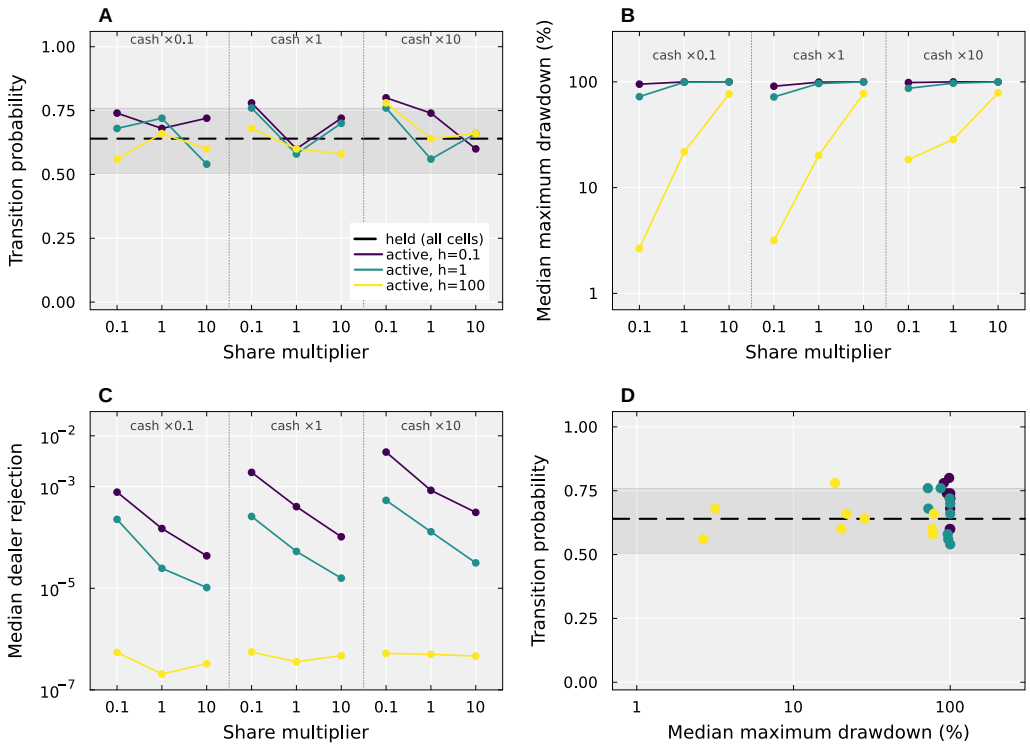


Figure 6: Changing trader cash and shares at the rule switch, with active technical traders, fraction $f = 0.18$, and 50 starting markets over 3,000 post-switch steps. In A–C, the nine resource combinations are grouped by cash multiplier, with share multiplier on the axis and a series for each dealer depth. **(A)** Majority probability. The dashed line and band give the held-condition probability and its 95% Wilson interval, identical across financial conditions. **(B)** Median maximum drawdown on a logarithmic axis. **(C)** Median fraction of post-switch eligible orders rejected, on a logarithmic axis. **(D)** Majority probability against median drawdown for all 27 active combinations, with the held reference repeated. The drawdown axis follows B. Probability axes cover the full possible range; all displayed drawdowns and rejection medians are strictly positive.

A Supplementary methods and validation

A.1 One complete simulation step

The simulation separated a desired decision from its execution. Orders used the decisions stored before the interior update, while technical traders used the new quoted price before the next interior decisions were calculated. A resource shortage could therefore stop an order without changing the state seen by neighboring traders. The uniform execution price depended on the net quantity supported by the entire batch (Eqs. 3 and 4), and differed from the quoted price left after settlement (Eq. 8). Executed counts specified the net transfer and its normalized imbalance (Eqs. 5 and 6), which changed the dealer reserves (Eq. 7). The new quoted price then updated the reference band (Eq. 10). Algorithms S1 and S2 specify this ordering for the main persistent-rule experiments with adaptive bands and no cost-basis restriction, including the transaction-fee extension. The pulse, alternative-rule, and frozen-band controls modify the corresponding updates as described in Appendix A.5.

Algorithm S1 One market step, $t-1 \to t$, in simulation order. Stored decisions generated orders. Clearing determined a common execution price, and the settled dealer reserves set the new quoted price.

Require: grid X_{t-1}; technical-trader decisions; trader books; dealer (W_{t-1}, Q_{t-1}); fee account G_{t-1} and rate τ; quoted price and band state $(P_{t-1}, \mu_{t-1}, \sigma_{t-1})$; rule field R_i; parameters; seed
Ensure: grid X_t; technical-trader decisions; settled books and dealer; quoted price and band state (P_t, μ_t, σ_t); order counts

if t is the scheduled rule-switch step **then**
  $R_i \leftarrow 19286$ for the preselected interior traders {preserve current decisions and financial state}
end if
if t is a scheduled resource intervention **then**
  for all traders i **do**
    $w_i \leftarrow g_w w_i$; $q_i \leftarrow \mathrm{round}(g_q q_i)$
  end for
end if
call Algorithm S2 using the stored decisions
receive settled books, dealer and fee account, order counts, base price F_t, and quoted price P_t
$\mu_t \leftarrow \mu_{t-1} + \alpha(P_t - \mu_{t-1})$
$\sigma_t \leftarrow \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1})$
if technical traders are active **then**
  for all technical traders i **do**
    $e_i(t) \leftarrow 0$ if $P_t < \mu_t - k_i\sigma_t$; 2 if $P_t > \mu_t + k_i\sigma_t$; 1 otherwise
  end for
else
  $e_i(t) \leftarrow 1$ for every technical trader i
end if
write $e_i(t)$ to the technical traders at selected edge positions; leave other edge positions in Hold
for all interior traders i, simultaneously **do**
  $c_i \leftarrow \sum_{j \in \partial i} s_j$ using old interior and new technical-trader decisions
  $s_i(t) \leftarrow \mathcal{R}_{R_i}(c_i)$
end for

Algorithm S2 Reserve clearing at a common base price without a cost-basis restriction. The binary search found the greatest supported integer net purchase in the direction of initial excess demand. The baseline uses $\tau = 0$.

Require: desired unit buys and sells from stored decisions; trader books $(w_i, q_i, \bar{c}_i)$; dealer (W, Q); fee account G; fee rate τ; previous quoted price; seed and step
Ensure: settled books and dealer; order counts; execution price and next quoted price
if dealer depth $h = 0$ **then**
  use the previous quoted price F; match buys with $w_i \geq (1+\tau)F$ and sells with $q_i \geq 1$
  apply priorities; use the buyer and seller cash, share, and cost-basis updates shown below
  update G as below; keep trading reserves at zero and the quoted price fixed
  return settled books, fee account, and order counts
end if
define $F(x) = W/(Q - x)$ for integer x satisfying $Q - x \geq 1$
define $B(x)$ as desired buyers with $w_i \geq (1+\tau)F(x)$ and $S(x)$ as desired sellers with $q_i \geq 1$
$d \leftarrow \operatorname{sign}[B(0) - S(0)]$; $\ell \leftarrow 0$
$u \leftarrow \min(B(0) - S(0), Q - 1)$ if $d > 0$, otherwise $S(0) - B(0)$
while $\ell < u$ **do**
  $k \leftarrow \lfloor (\ell + u + 1)/2 \rfloor$
  if $d[B(dk) - S(dk)] \geq k$ **then**
    $\ell \leftarrow k$
  else
    $u \leftarrow k - 1$
  end if
end while
$x \leftarrow d\ell$; $F_t \leftarrow F(x)$
collect eligible sets $\mathcal{B}$ and $\mathcal{S}$ at F_t
$n^B \leftarrow \min(|\mathcal{B}|, |\mathcal{S}| + x)$; $n^S \leftarrow n^B - x$
suppression $\leftarrow$ desired counts minus eligible counts
rejection $\leftarrow$ eligible counts minus executed counts
rank any rationed side by $z(\text{seed}, t, i, \text{side})$, breaking ties by identity
for all selected buyers i **do**
  $\bar{c}_i \leftarrow [\bar{c}_i q_i + (1+\tau)F_t]/(q_i + 1)$; $q_i \leftarrow q_i + 1$; $w_i \leftarrow w_i - (1+\tau)F_t$
end for
for all selected sellers i **do**
  $q_i \leftarrow q_i - 1$; $w_i \leftarrow w_i + (1-\tau)F_t$
end for
$Q' \leftarrow Q - x$; $W' \leftarrow WQ/Q'$; $P_t \leftarrow W'/Q'$
$G' \leftarrow G + \tau F_t(n^B + n^S)$
return settled books, dealer, fee account, counts, F_t, and P_t

A.2 Rationing and paired reproducibility

Each eligible order on a rationed side received a 64-bit score from a SplitMix64 finalizer applied to the simulation seed, step, trader identity, and order side. Sorting by score, then identity, made rationing reproducible while allowing priority to change over time. Paired markets used the same score for the same trader and step even after their eligible sets diverged. Scoring consumed no draws from the generator used for initialization. For paired rule switches, comparisons checked the pre-switch grid and Sell history, technical-trader coordinates and multipliers, and price history through the first post-switch quoted price. A pure rule switch preserved that first quoted price because the stored decisions still generated the same orders. A one-time Sell pulse and a resource intervention could change the first batch and therefore were required to agree only before the intervention. For held technical traders, resource comparisons also required identical Sell histories and final grids across all cash and share multipliers.

A.3 Conservation and limiting cases

Let total trader cash and shares be the sums over all traders. Outside scheduled resource interventions, their combination with dealer trading reserves and the fee account satisfied the conservation identities:

$$\sum_i w_i(t) + W_t + G_t = \sum_i w_i(0) + W_0 + G_0,$$
$$\sum_i q_i(t) + Q_t = \sum_i q_i(0) + Q_0.$$

The production wrapper checked these identities separately before and after an intentional resource change. Share conservation was exact. The allowed absolute cash discrepancy was the larger of 10^{-5} and 10^{-11} times total cash at the beginning of the checked interval. For positive dealer depth, checks required a finite positive quoted price, nonnegative trader resources, preservation of the reserve product to relative 10^{-9}, and agreement between the quoted price and reserve ratio. The implementation evaluated new dealer cash as WQ/Q' to limit accumulated rounding error; the equivalent transfer was $F_t x_t$. These checks ran after every newly completed simulation, before its execution record was written. The original experiments used $G_t = 0$. The fee extension included cumulative dealer revenue in cash conservation and checked that revenue against fees on completed orders. Fees were excluded from the reserve-product and quoted-price checks.

Zero dealer depth permitted only matched trader orders and kept price fixed. Technical traders then remained in Hold under either technical-trader condition, providing an exact comparison between active and held interior decisions. At positive dealer depth, tests checked settlement at the selected price, affordability after clearing, discrete rationing, and isolated round trips that restored dealer inventory and trading cash. With a positive fee, the isolated round trip also transferred cash from the trader to the dealer's fee account. The fee tests checked zero-fee agreement with the frozen original engine, affordability including fees, conservation, balanced orders, and the fixed-price zero-reserve case. They also covered a delayed rule switch, preservation of a common prehistory, and the held-decision invariance across financial interventions. An independent implementation of the held decision update served as a test oracle. Every production condition, including held decisions, was nevertheless rerun with the full financial simulator. Finite dealer depths were distinct model conditions: no finite-depth path was assumed to equal an unlimited dealer.

A.4 Simulation design and reproducibility

The clean rebuild began with an empty data and figure tree and a frozen copy of the corrected simulator, drivers, parameter manifest, and analysis code. No earlier simulation output or checkpoint was imported. Completed simulations wrote an execution ledger, while phase manifests and source hashes constrained any resumption to outputs created within this rebuild. All retained experiments were rerun, including held conditions whose decisions could be predicted without calculating finances. Seeds were retained to reproduce the planned starting markets; this was a complete recomputation under the corrected model, not an independent sample of starting markets. The corrected model replaced settlement at a standing quote followed by a separate exponential price update. That earlier mechanism admitted profitable isolated round trips, and its financial outputs did not enter this manuscript. Reserve clearing changed both settlement and price formation, so differences from the earlier model cannot be attributed to one parameter alone.

The four main campaigns comprised 30,550 simulations over 50 starting markets per condition and 3,000 post-switch steps (Supplementary Table S1). The selected paired-switch campaign used grid size $L = 64$, dealer depth $h = 100$, and active technical traders at fractions 0.10, 0.15, 0.20, and 0.25. Dealer comparisons used fractions 0.10, 0.15, 0.175, 0.185, 0.190, 0.195, 0.200, 0.205, 0.210, 0.225, and 0.250 at dealer depths $h = 0, 0.1, 0.5, 1, 2, 5, 10, 100$ with technical traders active and held. A newly simulated no-switch path was shared across fractions within each dealer depth, technical-trader condition, and starting market, avoiding duplicates within the same campaign. The size campaign used grid sizes $L = 32, 48, 64, 96, 128$ at fractions 0.140, 0.150, 0.160, 0.170, 0.175, 0.180, 0.185, 0.190, 0.200, 0.210, 0.225, and 0.250 before refinement. Its conditions were technical traders held at dealer depth $h = 100$ and technical traders active at $h = 1$ and $h = 100$. The cash and share campaign combined independent multipliers 0.1, 1, and 10 with fractions 0.170, 0.175, 0.180, and 0.185, dealer depths 0.1, 1, and 100, and both technical-trader conditions. Dealer resources increased with the trader count; no size series with fixed absolute dealer resources was included.

The supplementary controls retained these starting-market conventions while varying the intervention, dealer depth, band adaptation, or observation period. Their fixed design preceded this clean rebuild; adaptive refinement and example selection followed the numerical rules specified above. We used Julia 1.12.7 with the repository environment manifest, twelve simulation threads, and one garbage-collection thread. A seeded Xoshiro generator supplied setup draws [51]. Checkpointed drivers wrote atomic records and reconstructed a deterministic row order. A recorded scheduling-only revision assigned supplementary jobs to available workers individually. Validation against serial execution and the original schedule reproduced complete market histories exactly; the simulation equations, parameters, and analysis were unchanged. Python performed the bootstrap analysis and generated tables from the new summaries. The accompanying repository snapshot contains source hashes, parameter manifests, execution records, package versions, and derived outputs. The project repository is `https://github.com/varnerlab/Wolfram-markets-abm-paper`. The repository retains earlier implementations and outputs as a record of model development, outside the rebuild's permitted data inputs.

The example trajectories used newly computed 3,000-step summaries to select typical markets before inspecting their images. We reran the five selected conditions and all 50 starting markets at fraction $f = 0.150$ for 10,000 post-switch steps, each with a matched no-switch path (Fig. 4; Supplementary Table S11). Selection was not repeated using the longer histories. Comparisons at the original endpoint checked reproduction of the new short-run summaries; additional figure reruns checked full histories and final grids for the selected cases. The longer runs extended observations of the same starting markets and did not supply independent replications. Majority onset began with the first updated state after the rule switch, and all reported execution counts and dealer-rejection fractions used post-switch steps. Keeping these measurement windows common avoided mixing baseline trading into one campaign's financial summaries but not another's.

The completed rebuild was checked against its fixed parameter manifest and its execution ledger. All fixed conditions were present, and the new size and horizon campaigns with technical traders held agreed for 1,800 overlapping starting-market and fraction combinations. Across the main dealer, size, and resource campaign summaries, the largest absolute cash discrepancy was 3.40×10^{-6}, and share discrepancies were zero. These numerical checks accompanied the per-simulation reserve and conservation checks described above.

A.5 Additional controls and sensitivity analyses

The supplementary controls used the same initialization and baseline as the main comparisons, with grid size $L = 64$, dealer depth $h = 100$, and 3,000 post-intervention steps unless otherwise specified. Every switched run had a newly simulated no-switch control. Held conditions calculated full financial histories as well as decisions. A Sell pulse wrote Sell at the selected interior traders immediately before the first post-baseline orders, while retaining the baseline rule for later updates. Traders already in Sell were unchanged. A persistent intervention instead changed the selected rule without overwriting decisions, so its first price remained identical to the no-switch price. These paired interventions compared the same fractions and starting markets under both technical-trader conditions (Supplementary Table S5). The rule screen tested all eighteen ways of changing one response in the baseline lookup table, using twenty starting markets per condition and three fractions. Each screen entry was descriptive rather than a separate confirmatory hypothesis test (Supplementary Table S6).

The band control froze the recent-departure scale at its value immediately before the rule switch, while each technical trader retained its multiplier and the reference price continued to update. Adaptive and frozen paths shared their complete pre-switch histories. The frozen widths could therefore differ across dealer depths and starting markets. At fraction $f = 0.175$, the estimated majority-probability difference between dealer depths $h = 0.1$ and $h = 100$ was -0.06 with adaptive widths and -0.26 with frozen widths. Subtracting those two differences gave 0.20, with a 95% whole-market bootstrap interval of [-0.08, 0.50] (Supplementary Table S7). This comparison changed width adaptation while retaining the response to departures from a moving reference price. It therefore tested one part of the price feedback rather than removing technical trading.

Additional comparisons used dealer depths $h = 40, 100, 400$ at fraction $f = 0.20$ with active technical traders (Supplementary Table S8). For the reserve curve, differentiating log price with respect to normalized net trader purchases at initialization gives $2N_{\mathrm{tr}}/H$, approximately $2/h$. The chosen depths therefore spanned initial slopes 0.05, 0.02, and 0.005. No independent exponential

impact coefficient was varied: changing depth also changed dealer funding. These comparisons replaced an impact-coefficient sweep belonging to the superseded standing-quote model. They measured sensitivity within the corrected mechanism and did not hold available reserves fixed while changing price sensitivity. Neither the quoted price nor the length of a step was calibrated to an observed market.

The horizon campaign used fifty starting markets at fractions 0.140, 0.145, through 0.250, extending to 0.300 for grid size $L = 32$ to bracket short-horizon responses. Held technical traders prevented financial feedback throughout. Endpoint comparisons used the original 3,000 steps and size-dependent observation periods $3000(L/64)^z$ for $z = 1, 2$, alongside longer observations. Majority criteria of 0.4, 0.5, and 0.6 were extracted from each history. Interpolation and bootstrap intervals were calculated separately for every size, horizon, and criterion (Supplementary Table S9). The no-switch extension followed ten starting markets per technical-trader condition for 100,000 additional steps. The observed outcomes were active technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]; held technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]. These extensions reused starting markets and did not constitute independent replications. The observed frequencies of Sell majorities described the stated interval without establishing behavior at indefinitely long times.

We also characterized the shared pre-switch histories at dealer depth $h = 100$ with active technical traders. Across the fifty starting markets, mean interior Sell fractions were 0.2494 over steps 501–750 and 0.2504 over steps 751–1,000. In the latter interval, the median within-market log-return standard deviation was 0.000323, and median within-market first and ninety-ninth percentiles were -0.000730 and 0.000729. The mean fraction of exactly zero computed log returns was 0.007. These summaries described the starting markets without establishing stationarity or matching a calibrated empirical return distribution. The no-switch controls separately measured post-switch baseline financial variation across dealer depths (Supplementary Table S2).

The refined finite-size analysis resampled entire starting markets across fractions in 2,000 bootstrap replicates. The empirical estimate used the first upward crossing and allowed local decreases in probability, while the logistic fit imposed a smooth increasing response (Supplementary Table S10). Each replicate reselected the susceptibility maximum over the sampled grid. Consequently, intervals included variation in which sampled fraction supplied that maximum, but not interpolation bias, uncertainty from selecting the refinement grid, or changes from observing for longer. The recorded valid-bootstrap fraction identified samples in which the requested crossings remained bracketed. No susceptibility exponent or infinite-size limit was fitted.

A.6 Transaction-fee comparison and validation

The transaction-fee extension retained the original executed study and used a separate versioned simulation snapshot. Its manifest specified all 8,400 simulations before execution. The run checked cash conservation including the dealer's fee account, share conservation, nonnegative final trader resources, positive finite prices, the constant product of trading reserves, and the quoted price as their ratio. Dealer revenue was also checked against fees on every completed buy and sale. Each switched run shared its pre-switch financial and decision history with its no-switch control, including the first post-switch price and fee collection. With technical traders held, the complete Sell-fraction histories and final grids agreed exactly across fee rates and dealer depths. Separate tests checked zero-fee equivalence to the frozen original simulator, exhaustive feasible-quantity selection, fee-inclusive affordability, isolated round trips, balanced orders, and the zero-depth fixed-price case. Dealer fees were not available to fund trading in any condition. The additional source snapshot, manifest, per-run records, summaries, and validation report are stored in `output/rebuilt/transaction-fees-v1/`. The manuscript additions were generated from its validated summaries by `code/fees/WritePaper.py`. The fee study varied neither market size nor trader endowments and did not extend the observation period.